

PRIMARY 6 MATHEMATICS SCHEME OF WORK

FIRST TERM

Week	Topic	Content
1	Whole Numbers I	Place value up to millions and billions; Reading and writing large numbers; Comparing and ordering whole numbers; Rounding off to nearest 10, 100, 1000; Roman numerals up to 1000
2	Whole Numbers II & Basic Operations	Addition and subtraction of large numbers; Word problems involving whole numbers; Estimation and approximation; Properties of operations - commutative, associative, identity; Mental mathematics strategies
3	Multiplication and Division	Multiplication of 3-4 digit numbers; Long division; Division with remainders; Word problems; Multiples and factors; Prime and composite numbers; HCF and LCM
4	Order of Operations & Indices (Powers)	BODMAS/PEMDAS rule - Brackets, Orders, Division/Multiplication, Addition/Subtraction; Solving mixed operations; Introduction to indices - meaning of powers (e.g., 2^3 , 5^2); Laws of indices basics; Squares and square roots; Cubes and cube roots
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Fractions I	Types of fractions - proper, improper, mixed numbers; Equivalent fractions; Simplifying fractions; Comparing and ordering fractions; Converting between improper fractions and mixed numbers

8	Fractions II	Addition and subtraction of fractions with different denominators; Multiplication of fractions; Division of fractions; Word problems involving fractions; Fractions of quantities
9	Ratios and Proportions I	Meaning of ratio; Writing ratios in different forms; Simplifying ratios; Equivalent ratios; Sharing in given ratios; Ratio of family size and resources (practical applications); Comparing quantities using ratios
10	Ratios and Proportions II	Direct proportion; Inverse proportion; Ratio of two populations; Ratio of prevalence of HIV/AIDS between two sexes, two states (demography); Real-life applications of ratios; Word problems
11	REVISION	Review of all topics covered; Practice exercises; Problem-solving strategies; Question and answer sessions; Past questions practice
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 6 MATHEMATICS LESSON NOTES

FIRST TERM

WEEK 1: WHOLE NUMBERS I

OBJECTIVES By the end of the lesson, pupils should be able to:

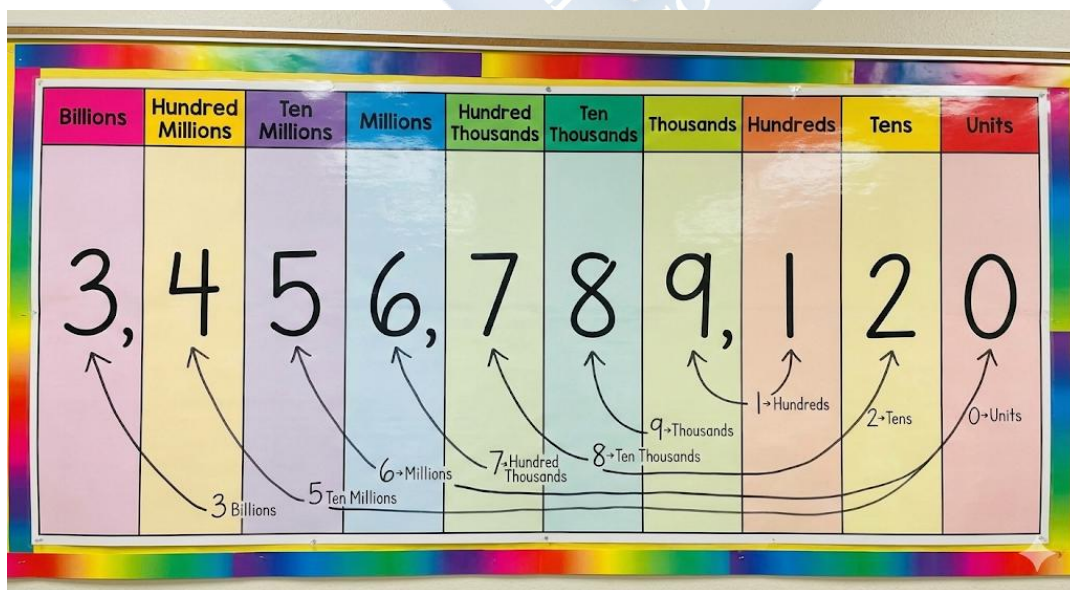
1. Read and write large whole numbers in figures and words (up to billions).
2. Identify the place value of digits in large numbers.
3. Compare and order whole numbers using symbols ($<$, $>$, $=$).
4. Read and write Roman Numerals up to 1000 (M).

CONTENT

1. Place Value up to Billions

Every digit in a number has a special position or "house" called its place value. This tells us the real worth or value of that digit. We read numbers in groups of three, separated by commas. These groups are Units, Thousands, Millions, and Billions.

Think of it like a family: The Units are the smallest (ones, tens, hundreds). The Thousands family is bigger (thousands, ten thousands, hundred thousands). The Millions and Billions families are even bigger!



- **Example:** In the number 3,456,789,120:
 - The digit 3 is in the Billions house. It means 3,000,000,000 (three billion).
 - The digit 4 is in the Hundred Millions house. It means 400,000,000.
 - The digit 5 is in the Ten Millions house. It means 50,000,000.
 - The digit 6 is in the Millions house. It means 6,000,000.
 - The 789 is in the Thousands family: 7 (Hundred Thousands), 8 (Ten Thousands), 9 (Thousands).
 - The 120 is in the Units family: 1 (Hundreds), 2 (Tens), 0 (Units).

2. Reading and Writing Large Numbers

To read a large number, read the number in each group, then say the family name only for the Thousands, Millions, and Billions groups. We don't say "units".

- **Example:** 2,500,000,000
 - Read the first group: Two
 - Say the family name: Billion
 - Read the next group: Five hundred million.
 - **Full reading:** Two billion, five hundred million.
- **Example:** 6,704,312 (Think: 6, 704, 312)
 - Read first group: Six + family name Million.
 - Read second group: Seven hundred and four + family name Thousand.
 - Read last group: Three hundred and twelve.
 - **Full reading:** Six million, seven hundred and four thousand, three hundred and twelve.

6,704,312

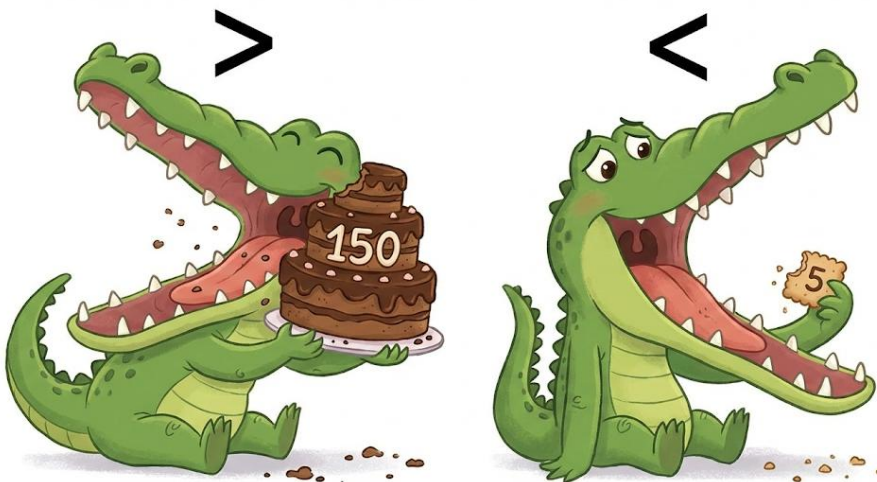
Six million,
seven hundred
and four
thousand,
three hundred
and twelve

3. Comparing and Ordering Numbers

To compare two numbers, start from the leftmost digit (the biggest place value). Find the first place where the digits are different. That tells you which number is bigger.

- **Symbols:**

- $>$ means **Greater Than** (The big mouth eats the bigger number). Example: $8 > 3$.
- $<$ means **Less Than** (The small point faces the smaller number). Example: $3 < 8$.
- $=$ means **Equal To**.



- **Example:** Compare 5,400,000 and 5,399,999.
 - Millions place: Both have 5 million.
 - Hundred thousands place: First number has 4, second has 3.
 - Since 4 is greater than 3, we stop. $5,400,000 > 5,399,999$.

4. Roman Numerals

Romans used letters from their alphabet to write numbers. We still use them for clocks, book chapters, and dates.

- **Basic Symbols:** I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000.

Roman Numeral	Value
I	1
V	5
X	10
L	50
C	100
D	500
M	1000

The Two Main Rules:

- **ADDITION RULE:** If a smaller numeral is written **AFTER** a larger one, you **ADD** them.
 - $VI = 5 (V) + 1 (I) = 6$.
 - $XIII = 10 (X) + 1 (I) + 1 (I) + 1 (I) = 13$.
 - $LX = 50 (L) + 10 (X) = 60$.
- **SUBTRACTION RULE:** If a smaller numeral is written **BEFORE** a larger one, you **SUBTRACT** it.

- $IV = 5 (V) - 1 (I) = 4$. (You cannot write four as IIII).
- $IX = 10 (X) - 1 (I) = 9$.
- $XC = 100 (C) - 10 (X) = 90$.
- **Example:** $MCC = 1000 (M) + 100 (C) + 100 (C) = 1200$.
- **Example:** $CM = 1000 (M) - 100 (C) = 900$.

EVALUATION

1. Write "Five billion, six hundred million" in figures.
2. In the number 4,782,100, what is the value of the digit 7? (i.e., 7 is worth how much?)
3. Write 2,400,000 in words.
4. Use $<$ or $>$ to make this true: 8,999,999 _____ 9,000,000.
5. Write the Roman Numeral for 500.
6. Write the Roman Numeral for 90. (Hint: Use the subtraction rule).
7. What number does MCX represent? Show your working.

ASSIGNMENT

1. **(Oral/Written)** Tell your parent or guardian the population of Nigeria (approx. 218 million). Ask them to say it, then you write it in figures.
 2. **(Drawing/Practical)** Draw a simple place value chart from Hundreds to Ten Millions. Write the number 34,567,890 in it and color the Ten Thousands digit.
 3. **(Written)** Write the year you were born and the current year in Roman Numerals.
-

WEEK 2: WHOLE NUMBERS II & BASIC OPERATIONS

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Add and subtract large numbers correctly.
2. Solve word problems involving addition and subtraction.
3. Round off numbers to the nearest 10, 100, and 1000.
4. Identify the commutative and associative properties of numbers.

CONTENT

1. Addition and Subtraction of Large Numbers

The key is to line up the numbers correctly by their place value. Write units under units, tens under tens, etc.

- **Addition (Carrying):** Add each column from right to left (units first). If the sum in a column is 10 or more, write the unit digit below and "carry" the tens digit to the next column on the left.
- **Subtraction (Borrowing):** Subtract each column from right to left. If the top digit is smaller than the bottom digit, "borrow" 1 from the next column on the left. This turns the top digit into a larger value.

A handwritten subtraction problem on a grey background. The problem is $50312 - 2457$. The result 2575 is written below a horizontal line. Red annotations show the borrowing process: a '4' is written above the first '5' and crossed out; a '9' is written above the '0' and crossed out; a '12' is written above the '3' and crossed out. Red arrows indicate the flow of borrowing from left to right: from the '5' to the '0', from the '9' to the '3', and from the '12' to the '1'.

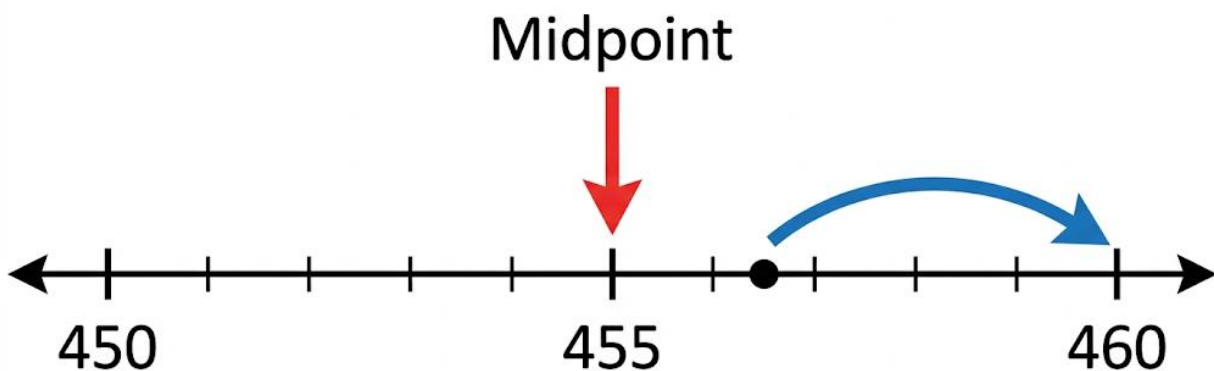
- **Real-life example:** If Lagos has about 15,000,000 people and Kano has about 4,000,000 people, what is their total population? $15,000,000 + 4,000,000 = 19,000,000$.

2. Estimation (Rounding Off)

Rounding helps us make numbers simpler and easier to work with when we don't need an exact answer (like estimating cost).

The Rule: Look at the digit to the right of the place value you are rounding to.

- **If it is 0, 1, 2, 3, or 4 (The "Weak" Numbers):** The digit you are rounding stays the same. Change all digits to the right to zero.
- **If it is 5, 6, 7, 8, or 9 (The "Strong" Numbers):** The digit you are rounding goes up by 1. Change all digits to the right to zero.



To round 456 to the nearest 10, see it's closer to 460 than 450, so round UP.

- **Round to the nearest 10:** Look at the units digit.
 - Round 456: The units digit is 6 (Strong). Round the 5 (tens) up to 6. Answer: 460.
- **Round to the nearest 100:** Look at the tens digit.
 - Round 4,230: The tens digit is 3 (Weak). The 2 (hundreds) stays 2. Answer: 4,200.
- **Round to the nearest 1000:** Look at the hundreds digit.
 - Round 4,821: The hundreds digit is 8 (Strong). Round the 4 (thousands) up to 5. Answer: 5,000.

3. Properties of Operations

These are special rules numbers always follow.

- **Commutative Property (Swap Rule):** For addition and multiplication only, swapping the order of numbers does not change the answer.
 - Addition: $5 + 3 = 8$ and $3 + 5 = 8$. (True)
 - Multiplication: $4 \times 2 = 8$ and $2 \times 4 = 8$. (True)
 - **Does NOT work** for Subtraction: $5 - 3 = 2$, but $3 - 5$ is not 2.
- **Associative Property (Grouping Rule):** For addition and multiplication only, how you group numbers (with brackets) does not change the answer.
 - Addition: $(2 + 3) + 4 = 5 + 4 = 9$. And $2 + (3 + 4) = 2 + 7 = 9$. (True)
 - Think of it as sharing food: Whether you add the garri to the soup first, or the meat to the soup first, the total amount of food is the same.

EVALUATION

1. Calculate: $4,521,000 + 3,110,200$.
2. Find the difference: $8,000,000 - 2,450,000$.
3. Round 675 to the nearest ten.
4. Round 4,821 to the nearest thousand.
5. True or False? Changing the order in subtraction gives the same answer. Explain with an example.
6. A village has 12,500 men and 13,200 women. What is the total population? If 5,000 are children, how many adults are there?

ASSIGNMENT

1. **(Practical)** Estimate the total cost: If a bag of rice is about ₦45,500 and a carton of noodles is about ₦8,300, roughly how much do they cost together? (Round to nearest ₦1000 first).
2. **(Written)** Solve this story problem: The government built 540 schools last year and 320 schools this year. How many schools were built in total?
3. **(Oral/Written)** Tell your friend two examples of the Commutative Property using things in your home (e.g., 2 plates + 3 plates = 3 plates + 2 plates).



WEEK 3: MULTIPLICATION AND DIVISION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Multiply 3-digit and 4-digit numbers by 2-digit numbers.
2. Divide whole numbers by 2-digit numbers (Long Division).
3. Find the Highest Common Factor (HCF) of two numbers.
4. Find the Lowest Common Multiple (LCM) of two numbers.

CONTENT

1. Multiplication

When multiplying by a 2-digit number, multiply by the units digit first, then by the tens digit.

Remember to add a zero as a placeholder when you start multiplying by the tens digit.

- **Example:** 423×12
 - Multiply by 2 (units): $423 \times 2 = 846$.
 - Multiply by 1 (tens): $423 \times 10 = 4,230$. (Write this starting under the tens column, which means putting a 0 in the units column first).
 - Add the two results: $846 + 4,230 = 5,076$.

$$\begin{array}{r} 423 \\ \times 12 \\ \hline 846 \\ 0 \\ 4230 \\ \hline 5076 \end{array}$$

← Part 1: 423×2

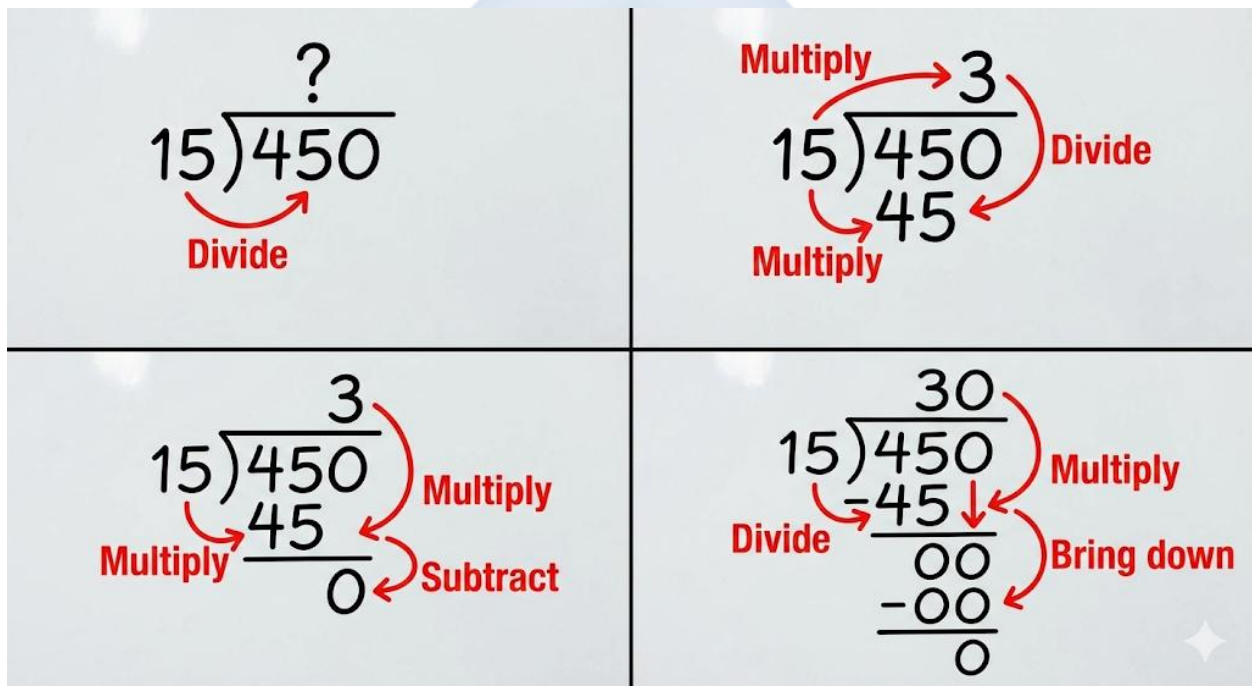
← Part 2: 423×10 (placeholder 0)

← Final Sum: $846 + 4230$

2. Long Division (The DMSB Method)

Use the steps: Divide, Multiply, Subtract, Bring down. Repeat until no digits remain to bring down.

- **Example:** $450 \div 15$
 - **D:** How many 15s are in 45? 3. Write 3 above the line.
 - **M:** Multiply $3 \times 15 = 45$. Write it below 45.
 - **S:** Subtract: $45 - 45 = 0$.
 - **B:** Bring down the next digit, which is 0.
 - Repeat: How many 15s in 0? 0. Write 0 above. Answer: 30.



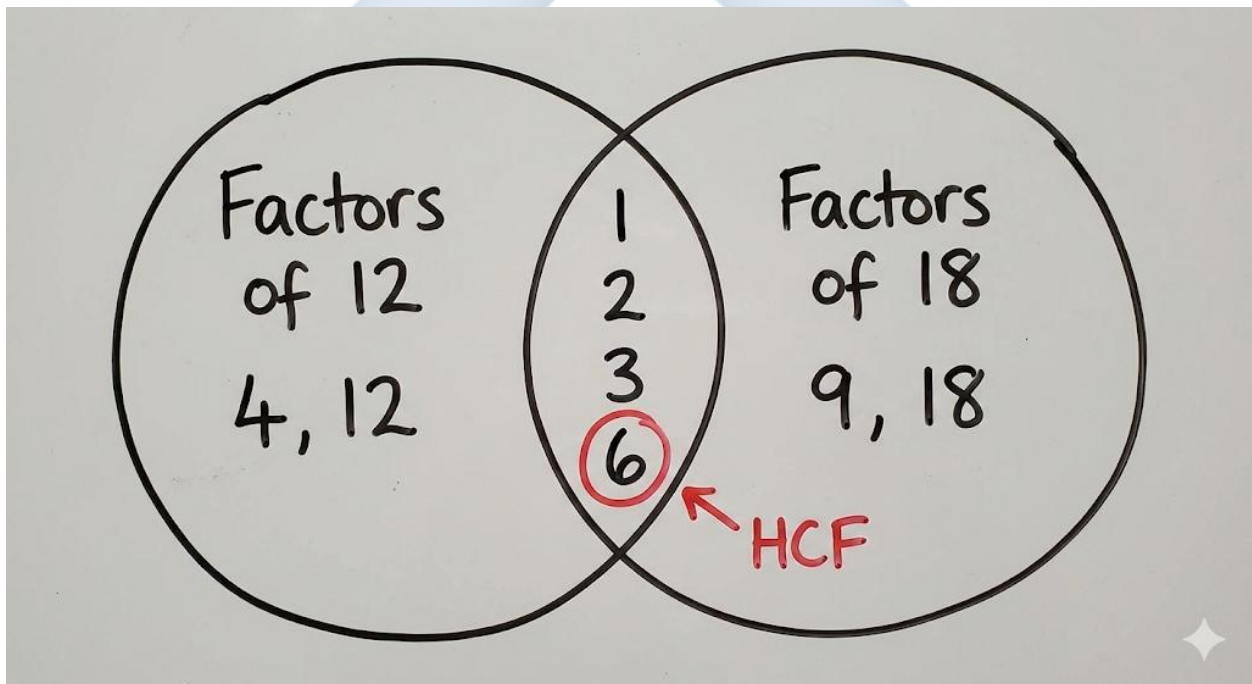
3. Factors and HCF

- **Factors:** Numbers that divide another number exactly, with no remainder.
 - Factors of 12: 1, 2, 3, 4, 6, 12. (Because $12 \div 1 = 12$, $12 \div 2 = 6$, etc.)
 - Factors of 18: 1, 2, 3, 6, 9, 18.
- **Highest Common Factor (HCF):** The largest factor that is common to (shared by) both numbers.

- Common factors of 12 and 18: 1, 2, 3, 6.
- The largest is 6. So, **HCF of 12 and 18 = 6.**

4. Multiples and LCM

- **Multiples:** The results you get when you multiply a number by 1, 2, 3, 4... (It's the number's "times table").
 - Multiples of 4: 4, 8, 12, 16, 20, 24...
 - Multiples of 6: 6, 12, 18, 24, 30...
- **Lowest Common Multiple (LCM):** The smallest multiple that is common to both numbers (the first number that appears in both lists).
 - Common multiples of 4 and 6: 12, 24, 36...
 - The first/smallest is 12. So, **LCM of 4 and 6 = 12.**



EVALUATION

1. Multiply: 125×14 .
2. Divide using long division: $625 \div 25$.
3. Write the first four multiples of 8.
4. List all the factors of 20.

5. Find the HCF of 10 and 15.
6. Find the LCM of 3 and 4. Explain how you know.

ASSIGNMENT

1. **(Written)** A bag contains 250 marbles. How many marbles are in 12 such bags?
 2. **(Practical/Drawing)** Find the HCF of 24 and 36 by listing their factors in two columns and circling the common ones.
 3. **(Oral/Written)** Two traffic lights blink. One blinks every 4 seconds, the other every 6 seconds. If they blink together now, after how many seconds will they blink together again? (This is the LCM).
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WEEK 4: ORDER OF OPERATIONS & INDICES (POWERS)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain the BODMAS rule.
2. Solve mathematical expressions using BODMAS.
3. Write numbers in index form (powers).
4. Calculate the square and square root of whole numbers.

CONTENT

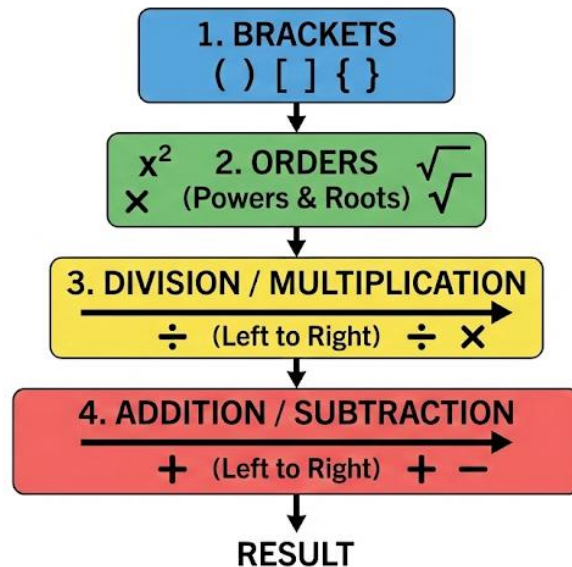
1. BODMAS Rule

When a mathematical expression has many different operations (+, -, $\times$, $\div$), we must solve it in a specific order to get the one correct answer. This order is remembered by the word **BODMAS**.

- **B – Brackets ()** : Solve anything inside brackets first.
- **O – Orders (Powers like 2 and square roots $\sqrt{}$)** : Solve powers or roots next.
- **D – Division ($\div$) M – Multiplication ($\times$)** : Do division and multiplication next, working from left to right.
- **A – Addition (+) S – Subtraction (-)** : Do addition and subtraction last, working from left to right.

Think of it as a school rule: You must visit the Brackets office first, then the Orders office, before you can do the DM or AS in the classroom.

ORDER OF OPERATIONS LADDER

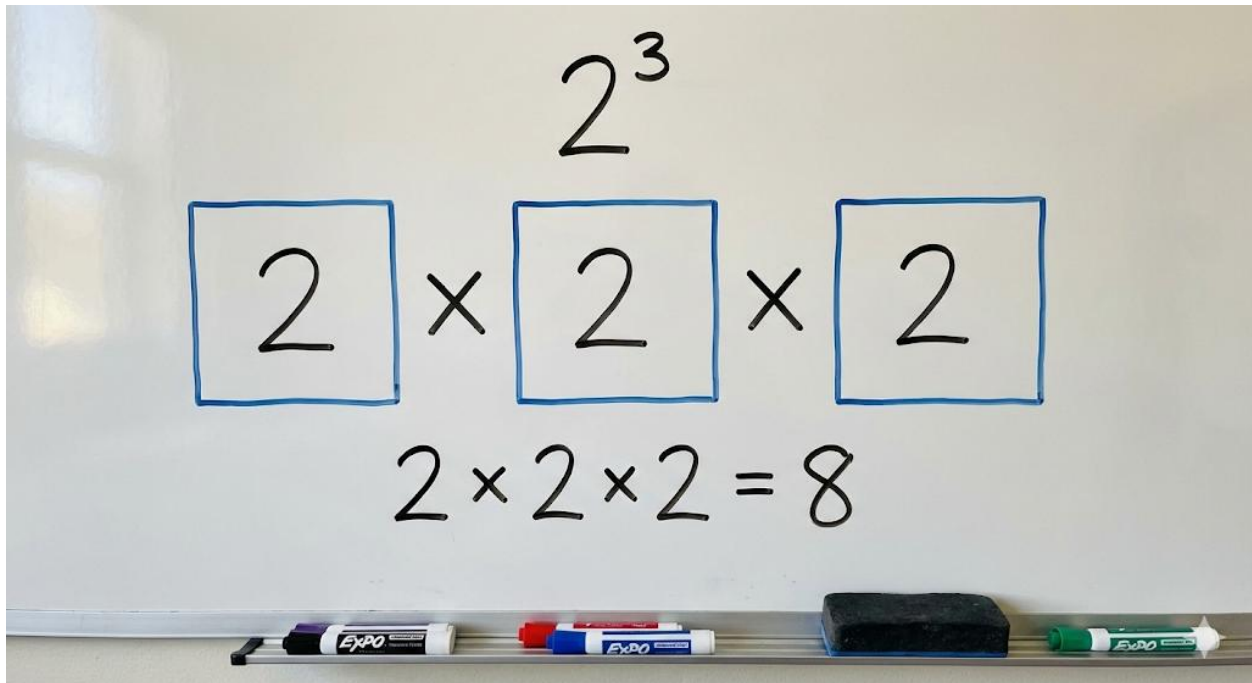


- **Example:** $5 + 2 \times 3$
 - Wrong way (doing + first): $5+2=7$, then $7 \times 3=21$.
 - **BODMAS way:** Multiplication comes before Addition. So, $2 \times 3 = 6$ first. Then, $5 + 6 = 11$. (Correct)

2. Indices (Powers)

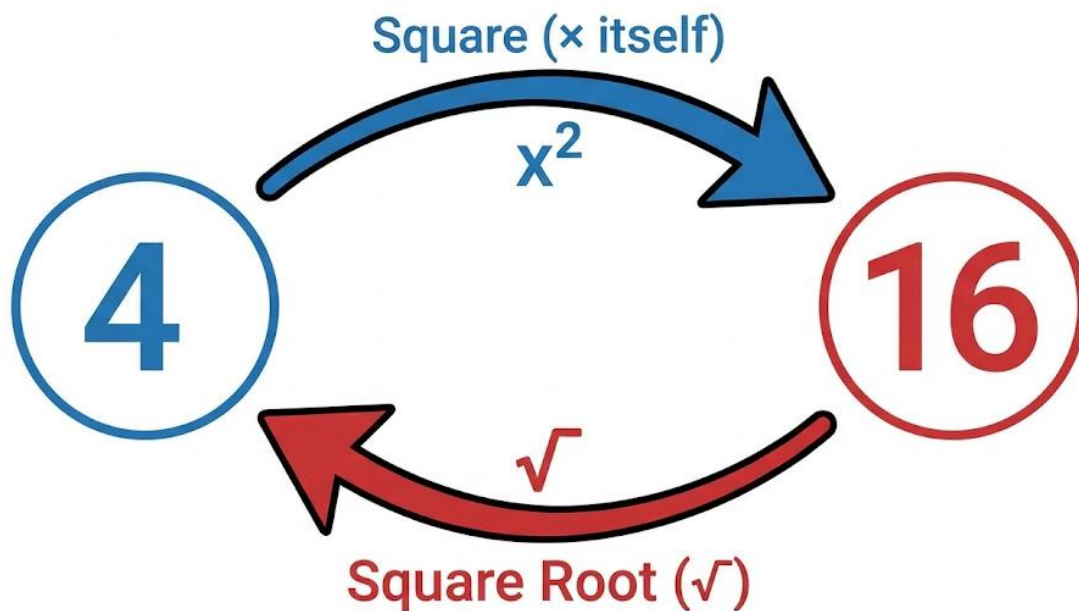
An index (plural: indices) or power tells us how many times a number (the base) is multiplied by itself. It is a short way to write long multiplications.

- The small raised number is the index or power.
- 2^3 means: 2 (the base) multiplied by itself 3 times. $2^3 = 2 \times 2 \times 2 = 8$.
 - It does **NOT** mean $2 \times 3 = 6$. This is a common mistake.
- 5^2 means: $5 \times 5 = 25$. We say "five squared".
- 4^5 means: $4 \times 4 \times 4 \times 4 \times 4$.



3. Squares and Square Roots

- **Squaring a Number:** This means multiplying a number by itself. The result is called a perfect square.
 - Square of 4 = $4^2 = 4 \times 4 = 16$. So, 16 is a perfect square.
 - Square of 10 = $10^2 = 10 \times 10 = 100$.
- **Square Root of a Number ($\sqrt{}$):** This is the opposite of squaring. It asks: "Which number, when multiplied by itself, gives this answer?"
 - $\sqrt{16} = ?$ (What number $\times$ itself = 16?) Answer: 4, because $4 \times 4 = 16$.
 - $\sqrt{100} = 10$.
 - $\sqrt{9} = 3$.



EVALUATION

1. Solve using BODMAS: $10 + 4 \times 2$.
2. Solve: $20 - (5 + 5)$.
3. Write $3 \times 3 \times 3 \times 3$ in index form.
4. What is the value of 2^4 ?
5. What is 7 squared (7^2)?
6. Find $\sqrt{81}$.

ASSIGNMENT

1. **(Written)** Solve: $15 \div 3 + 2 \times 4$. Show your BODMAS steps.
2. **(Practical/Drawing)** Draw 5 rows of 5 dots each. How many dots total? This shows 5^2 . Write the answer as a power.
3. **(Oral)** Explain to a classmate the difference between "3 squared" and "3 multiplied by 2".

WEEKS 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7: FRACTIONS I

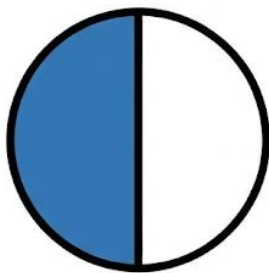
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Distinguish between proper, improper, and mixed fractions.
2. Convert improper fractions to mixed numbers and vice versa.
3. Simplify fractions to their lowest terms.
4. Compare fractions using LCM.

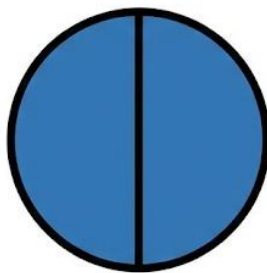
CONTENT

1. Types of Fractions

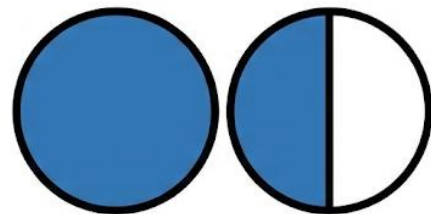
A fraction has two parts: **Numerator (top)** and **Denominator (bottom)**. The denominator tells how many equal parts the whole is split into. The numerator tells how many of those parts we have.



Proper: $\frac{1}{2}$



Improper: $\frac{2}{2}$



Mixed: $1\frac{1}{2}$

- **Proper Fraction:** The numerator is smaller than the denominator. The value is less than 1 whole. E.g., $\frac{1}{2}$, $\frac{3}{4}$, $\frac{2}{3}$. (You have a part of a whole).
- **Improper Fraction:** The numerator is equal to or larger than the denominator. The value is 1 or more. E.g., $\frac{5}{2}$, $\frac{7}{3}$, $\frac{4}{4}$. (You have one or more wholes, plus extra parts).

- **Mixed Number:** A combination of a whole number and a proper fraction. E.g., $2\frac{1}{2}$, $1\frac{3}{4}$. It shows whole things and a part of another.

2. Converting Fractions

- **Improper Fraction to Mixed Number:** Divide the numerator by the denominator.
 - **Example:** $7/2$
 - How many 2s are in 7? 3 times ($3 \times 2 = 6$).
 - Remainder is 1.
 - Answer: 3 wholes and 1 out of 2 parts = $3\frac{1}{2}$.
- **Mixed Number to Improper Fraction:** Multiply, Add, Keep.
 - **Example:** $1\frac{2}{3}$
 - Multiply the whole number by the denominator: $1 \times 3 = 3$.
 - Add the result to the numerator: $3 + 2 = 5$.
 - Keep the same denominator: $5/3$.

3. Equivalent Fractions & Simplifying

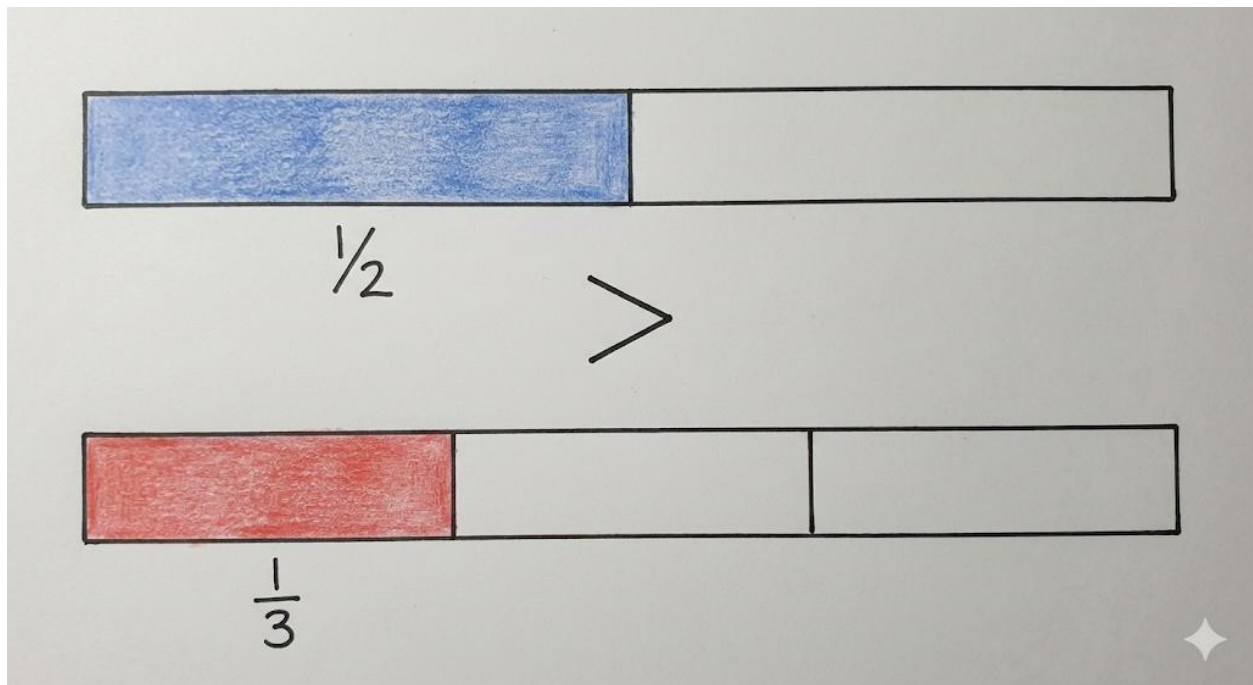
- **Equivalent Fractions:** Different fractions that represent the same value or portion. You get them by multiplying or dividing the numerator and denominator by the same number.
 - $\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{50}{100}$. Look at a half of a pie; it's the same size whether the pie is cut into 2, 4, or 6 pieces.
- **Simplifying (Lowest Terms):** We make a fraction as simple as possible by dividing the numerator and denominator by their Highest Common Factor (HCF).
 - Simplify $4/8$: The HCF of 4 and 8 is 4. Divide both by 4: $(4 \div 4)/(8 \div 4) = \frac{1}{2}$. This is the simplest form.

4. Comparing Fractions

To compare fractions with different denominators (like $\frac{1}{2}$ and $\frac{1}{3}$), make their denominators the same (find a common denominator).

- Compare $\frac{1}{2}$ and $\frac{1}{3}$.
 - Find the LCM of 2 and 3 = 6.

- Convert: $\frac{1}{2} = (1 \times 3)/(2 \times 3) = 3/6$. $\frac{1}{3} = (1 \times 2)/(3 \times 2) = 2/6$.
- Now compare $3/6$ and $2/6$. Since $3 > 2$, therefore $\frac{1}{2} > \frac{1}{3}$.



EVALUATION

1. Identify as Proper, Improper, or Mixed: a) $9/4$ b) $2/5$ c) $3\frac{1}{2}$.
2. Convert $5/2$ to a mixed number.
3. Convert $2\frac{1}{4}$ to an improper fraction.
4. Simplify $10/20$ to its lowest term.
5. Which is bigger: $\frac{1}{4}$ or $\frac{1}{2}$? Show your working.

ASSIGNMENT

1. **(Written)** Convert $15/4$ to a mixed number.
2. **(Practical/Drawing)** Draw two rectangles. Shade $12/16$ of the first. Then, simplify $12/16$ and shade the equivalent fraction in the second rectangle. What do you notice?
3. **(Oral/Written)** Arrange $\frac{1}{2}$, $\frac{1}{5}$, and $\frac{1}{3}$ in ascending order. Explain how you used a common denominator.

WEEK 8: FRACTIONS II (OPERATIONS)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Add and subtract fractions with different denominators.
2. Multiply fractions.
3. Divide fractions involving whole numbers and mixed numbers.
4. Calculate a fraction of a quantity.

CONTENT

1. Addition and Subtraction

You can only add or subtract fractions directly if they have the same denominator. If not, find a common denominator first.

- **Example:** $\frac{1}{2} + \frac{1}{3}$
 - Find Common Denominator: LCM of 2 and 3 is 6.
 - Convert: $\frac{1}{2} = \frac{3}{6}$, $\frac{1}{3} = \frac{2}{6}$.
 - Add Numerators: $3 + 2 = 5$.
 - Keep Denominator: **$\frac{5}{6}$** .
- Subtraction follows the same steps: Find common denominator, convert, subtract numerators.

2. Multiplication of Fractions

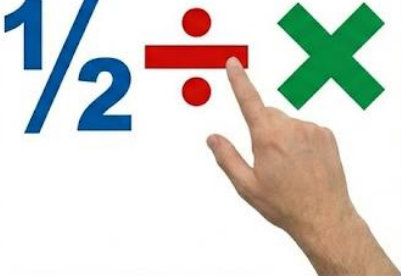
This is straightforward. Multiply the numerators together and the denominators together. Simplify if possible before multiplying.

- **Example:** $\frac{2}{3} \times \frac{4}{5} = (2 \times 4) / (3 \times 5) = \frac{8}{15}$.
- **Simplifying first:** $\frac{2}{3} \times \frac{9}{10}$. Notice 3 and 9 can be simplified ($3 \div 3 = 1$, $9 \div 3 = 3$). It becomes $\frac{2}{1} \times \frac{3}{10} = \frac{6}{10} = \frac{3}{5}$.

3. Division of Fractions

Use the **KCF** method: **K**eeP, **C**hange, **F**lip.

- **Keep** the first fraction.
- **Change** the $\div$ sign to a $\times$ sign.
- **Flip** the second fraction (swap its numerator and denominator).
- **Example:** $\frac{1}{2} \div \frac{1}{4}$
 - Keep: $\frac{1}{2}$
 - Change: $\div$ becomes $\times$
 - Flip: $\frac{1}{4}$ becomes $\frac{4}{1}$
 - Multiply: $\frac{1}{2} \times \frac{4}{1} = \frac{4}{2} = 2$.

STEP 1: KEEP	STEP 2: CHANGE	STEP 3: FLIP
KEEP $\frac{1}{2}$ 	CHANGE $\div$ to $\times$ 	FLIP $\frac{1}{4}$ to $\frac{4}{1}$ 

4. Fraction of a Quantity

The word "**of**" in mathematics often means "**multiply**".

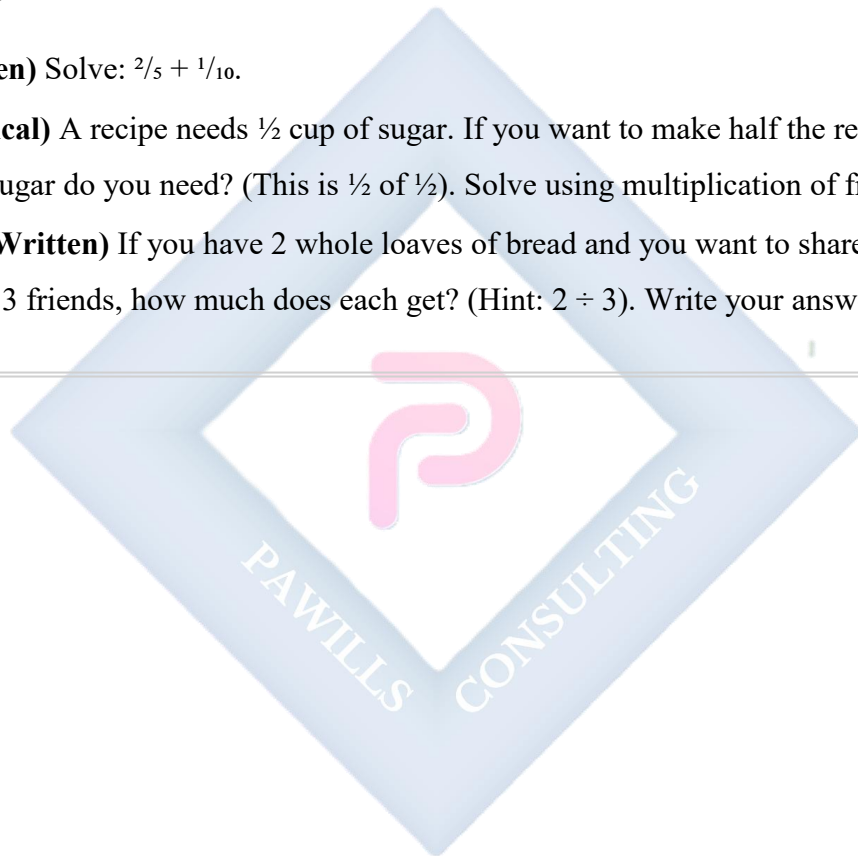
- To find $\frac{1}{2}$ of 10 oranges:
 - Write $\frac{1}{2} \times 10$. (Think of 10 as $\frac{10}{1}$).
 - Multiply: $(1 \times 10) / (2 \times 1) = \frac{10}{2} = 5$ oranges.
- Finding $\frac{3}{4}$ of 200 Naira:
 - $\frac{3}{4} \times 200 = (3 \times 200) / 4 = 600 / 4 = 150$ Naira.

EVALUATION

1. Solve: $\frac{1}{4} + \frac{1}{2}$.
2. Solve: $\frac{3}{4} - \frac{1}{4}$.
3. Multiply: $\frac{1}{3} \times \frac{3}{5}$.
4. Divide: $2 \div \frac{1}{2}$. (Hint: Write 2 as $\frac{2}{1}$ first).
5. Find $\frac{2}{3}$ of 12.

ASSIGNMENT

1. **(Written)** Solve: $\frac{2}{5} + \frac{1}{10}$.
2. **(Practical)** A recipe needs $\frac{1}{2}$ cup of sugar. If you want to make half the recipe, how much sugar do you need? (This is $\frac{1}{2}$ of $\frac{1}{2}$). Solve using multiplication of fractions.
3. **(Oral/Written)** If you have 2 whole loaves of bread and you want to share them equally among 3 friends, how much does each get? (Hint: $2 \div 3$). Write your answer as a fraction.



WEEK 9: RATIOS AND PROPORTIONS I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define ratio.
2. Write ratios in their simplest form.
3. Share a quantity in a given ratio.
4. Solve practical problems involving ratios.

CONTENT

1. Meaning of Ratio

A ratio is a way of comparing two or more quantities of the same kind. It tells us how much of one thing there is compared to another. We write ratios using a colon : (e.g., 2:3) and say "two is to three".

- **Example:** In a basket, there are 4 oranges and 5 mangoes.
 - The ratio of oranges to mangoes is **4:5**.
 - The ratio of mangoes to oranges is **5:4**.

2. Simplifying Ratios

Like fractions, ratios should be expressed in their simplest form. Divide both sides of the ratio by the same number (their HCF).

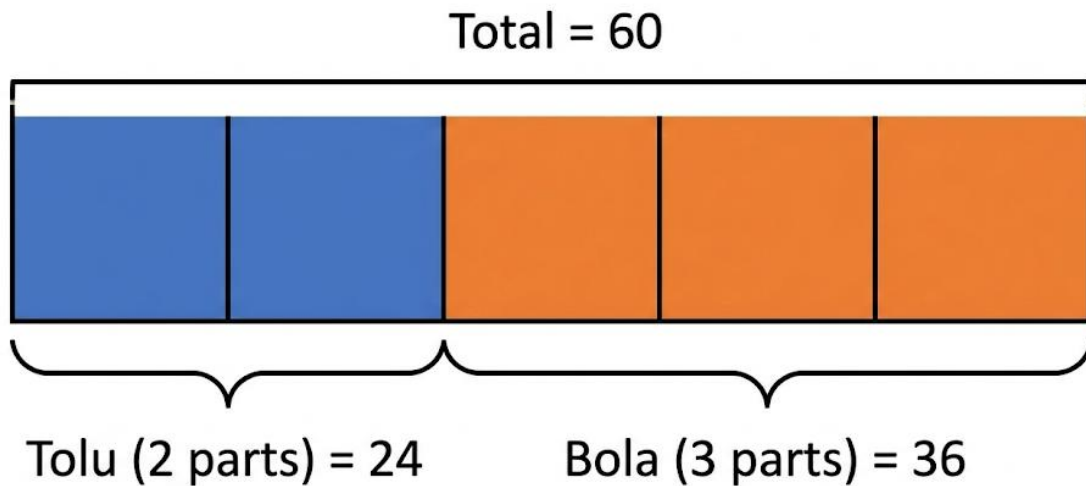
- **Example:** There are 10 boys and 20 girls in a group.
 - Ratio of boys to girls = 10:20.
 - HCF of 10 and 20 is 10.
 - Divide both sides by 10: $(10 \div 10) : (20 \div 10) = \mathbf{1:2}$.
 - This means for every 1 boy, there are 2 girls.

3. Sharing in a Given Ratio

This is a very important use of ratios. We share a total amount according to the ratio.

- **Example:** Share 60 oranges between Tolu and Bola in the ratio 2:3.

- Add the ratio parts to find the total parts: $2 + 3 = 5$ parts.
- Find the value of 1 part: Total quantity $\div$ Total parts = $60 \div 5 = \mathbf{12 \text{ oranges per part}}$.
- Multiply to find each share:
 - Tolu gets 2 parts: $2 \times 12 = \mathbf{24 \text{ oranges}}$.
 - Bola gets 3 parts: $3 \times 12 = \mathbf{36 \text{ oranges}}$.
- Check: $24 + 36 = 60$. Correct.



4. Ratio in Real Life (Family Resources)

Ratios help us understand real situations. If a family's monthly income is ₦50,000 and they have 5 children, the ratio of money to children is $50,000 : 5$, which simplifies to $10,000 : 1$. This is a simple way to see how resources are spread. A family with a higher income-per-child ratio (e.g., $₦20,000 : 1$) may find it easier to manage than one with a lower ratio.

EVALUATION

1. Write the ratio of 4 books to 8 books in its simplest form.
2. Express 50 kobo to 1 Naira as a ratio. (Hint: 1 Naira = 100 kobo).
3. Share 20 sweets between A and B in the ratio $1:1$. What is this type of sharing called?

4. Share 15 yams between two families in the ratio 1:2. How many yams does the family with the larger share get?

ASSIGNMENT

1. **(Written)** Simplify the ratio 15:25.
2. **(Practical)** Share 500 Naira between Musa and John in the ratio 2:3. Show your steps.
3. **(Oral/Investigation)** At home, find the ratio of spoons to plates in your kitchen. Write it and simplify it if possible.



WEEK 10: RATIOS AND PROPORTIONS II

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define Direct Proportion and Inverse Proportion.
2. Solve word problems on direct proportion.
3. Solve word problems on inverse proportion.
4. Apply ratios to real-life statistics.

CONTENT

1. Direct Proportion

Two quantities are in **direct proportion** if when one quantity increases, the other quantity increases at the same rate, and when one decreases, the other decreases at the same rate. Their ratio stays the same.

- **Real-life link:** The more sachets of water you buy, the more money you pay. Cost is directly proportional to number of sachets.
- **Solving using the Unitary Method:**
 - **Problem:** If 4 oranges cost ₦20, how much do 6 oranges cost?
 - Find cost of 1 orange (unit cost): $\text{₦}20 \div 4 = \text{₦}5$ per orange.
 - Find cost of 6 oranges: $\text{₦}5 \times 6 = \text{₦}30$.

Oranges	Cost
4	₦20
1	₦5
6	₦30

Diagram illustrating the Unitary Method for finding the cost of 6 oranges:

- From 4 oranges (₦20), divide by 4 to find the unit cost of 1 orange (₦5).
- From the unit cost of 1 orange (₦5), multiply by 6 to find the cost of 6 oranges (₦30).

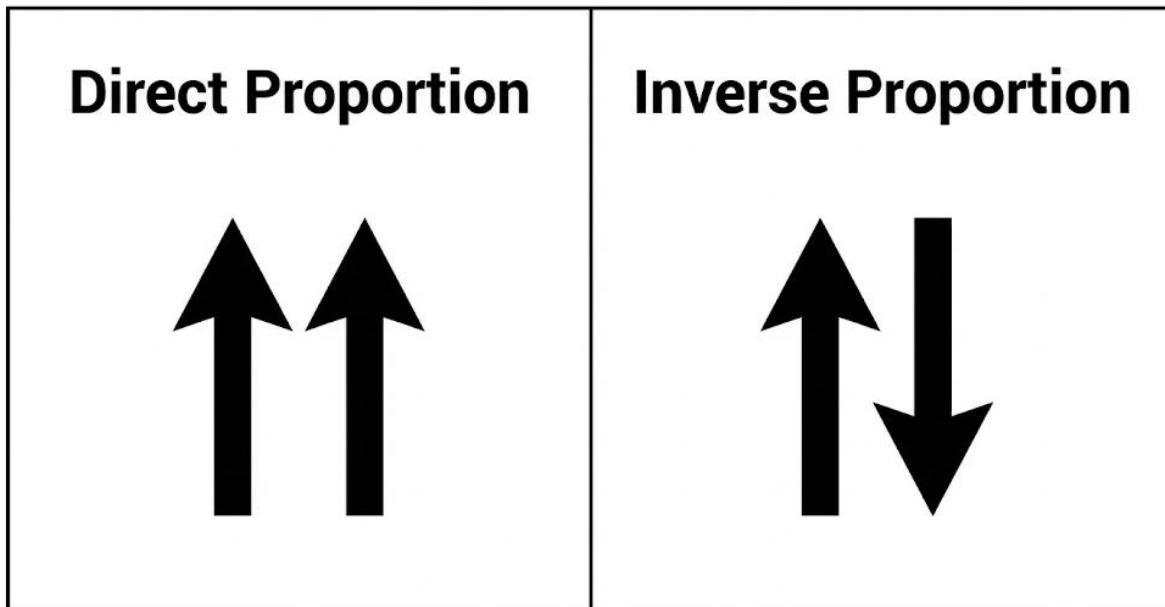
2. Inverse Proportion

Two quantities are in **inverse proportion** if when one quantity increases, the other quantity decreases, and vice-versa. Their product (multiplication result) stays constant.

- **Real-life link:** The more people working on a job, the less time it will take to finish.
- **Solving using the Constant Product Method:**
 - **Problem:** If 2 men can dig a trench in 4 days, how long will 4 men take?
 - Find the constant total work: $\text{Men} \times \text{Days} = \text{Work}$. $2 \text{ men} \times 4 \text{ days} = 8 \text{ "man-days"}$ of work.
 - Divide total work by the new number of men: $8 \text{ man-days} \div 4 \text{ men} = \mathbf{2 \text{ days}}$.

3. Distinguishing Direct and Inverse Proportion

- **Direct:** "The more... the more..." or "The less... the less...". (Cost & items, Distance & time at constant speed).
- **Inverse:** "The more... the less..." or "The less... the more...". (Workers & time, Speed & time for a fixed distance).



4. Application: Demographic Ratios

Ratios and proportions are used to understand important national statistics, like health or population data.

- **Example:** In a health survey of 1000 people, 50 men and 100 women were found to have a certain condition.
 - Ratio of affected men to women = $50:100 = 1:2$.
 - This simple ratio quickly shows that for every 1 affected man, there are 2 affected women. This helps the government know which group may need more attention in a health campaign.

EVALUATION

1. If 3 pens cost ₦150, what is the cost of 1 pen?
2. If 1 pen costs ₦50, what is the cost of 10 pens? (Is this direct or inverse?)
3. If 5 men finish a job in 2 days, how long will 1 man take? (First find the total "man-days" of work).
4. Identify as Direct or Inverse: "The more petrol I buy, the more money I pay."
5. Identify as Direct or Inverse: "The faster I run, the less time it takes to get to school."

ASSIGNMENT

1. **(Written)** If 10 bags of rice cost ₦50,000, find the cost of 3 bags using the unitary method.
 2. **(Practical Reasoning)** If 4 labourers can weed a farm in 6 hours, how many hours will 8 labourers take? Explain why your answer makes sense.
 3. **(Oral/Written)** Give one real-life example each of direct proportion and inverse proportion that you see around you.
-

PRIMARY 6 MATHEMATICS SCHEME OF WORK

SECOND TERM

Week	Topic	Content
1	Percentages I	Meaning of percentage; Converting fractions to percentages; Converting decimals to percentages; Converting percentages to fractions and decimals; Percentage of quantities
2	Percentages II	Calculating percentage increase and decrease; Profit and loss percentages; Discount and VAT; Simple interest; Ratio and percentage relationships; Word problems
3	Demography & Money I	Population - meaning and importance; Population growth and decline; Birth rate and death rate; Census; Population density; Economic consequences of population changes; Planning and resource allocation
4	Money II - Currency Conversion	Nigerian currency (Naira and Kobo); Foreign currencies (Dollar, Pound, Euro, etc.); Exchange rates; Converting between currencies; Buying and selling foreign currency; International trade basics; Travel money calculations
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Algebraic Processes - Open Sentences	Understanding variables; Simple equations with one unknown; Solving open sentences (e.g., $x + 5 = 12$); Using substitution; Forming simple equations from word problems; Introduction to algebraic expressions
8	Measurement - Length, Weight & Time I	Units of length - mm, cm, m, km; Converting between units; Measuring and estimating lengths; Units of weight - g, kg, tonne; Converting between units; Units of time - seconds, minutes, hours, days, weeks,

		months, years
9	Measurement - Time & Speed	Converting between time units; 12-hour and 24-hour clock; Time calculations - duration, timetables; Athletics time (recording race times); Speed - meaning and formula ($\text{Speed} = \text{Distance} \div \text{Time}$); Calculating speed, distance, and time; Average speed
10	Measurement - Perimeter, Area & Volume	Perimeter of rectangles, squares, triangles, and irregular shapes; Area of rectangles, squares, triangles, parallelograms, and circles; Volume of cubes and cuboids; Surface area basics; Capacity - litres, millilitres; Converting between capacity units; Relationship between volume and capacity
11	REVISION	Review of all topics covered; Practice exercises; Problem-solving strategies; Question and answer sessions; Past questions practice
12	EXAMINATION	End of term assessment
13	CLOSING	Report card distribution; Vacation assignment

PRIMARY 6 MATHEMATICS LESSON NOTES

SECOND TERM

WEEK 1: PERCENTAGES I

OBJECTIVES By the end of the lesson, pupils should be able to:

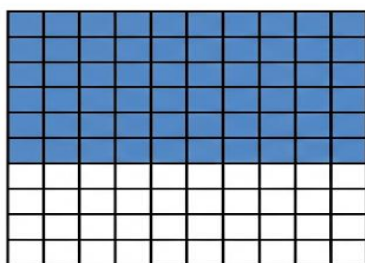
1. Define percentage as a fraction with a denominator of 100.
2. Convert fractions and decimals to percentages.
3. Convert percentages to fractions (in lowest terms) and decimals.
4. Calculate a given percentage of a quantity.

CONTENT

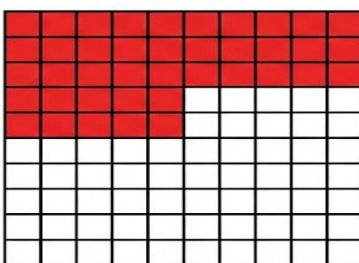
1. Meaning of Percentage

The word "percent" comes from Latin words "per centum", meaning "out of one hundred" or "for every hundred". The symbol for percentage is %. It is a special way of writing a fraction with a denominator of 100.

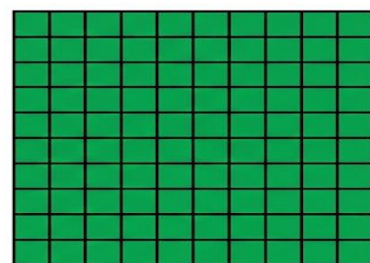
- Think of a test score: If you score 80%, it means you got 80 correct answers out of every 100 questions. If you get 50% in a test, you got half of the answers right, because 50 out of 100 is the same as $\frac{1}{2}$.
- In the market, a 25% discount means you will pay N25 less for every N100 of the original price.



50% or $\frac{1}{2}$



25% or $\frac{1}{4}$



80%

2. Converting Fractions to Percentages

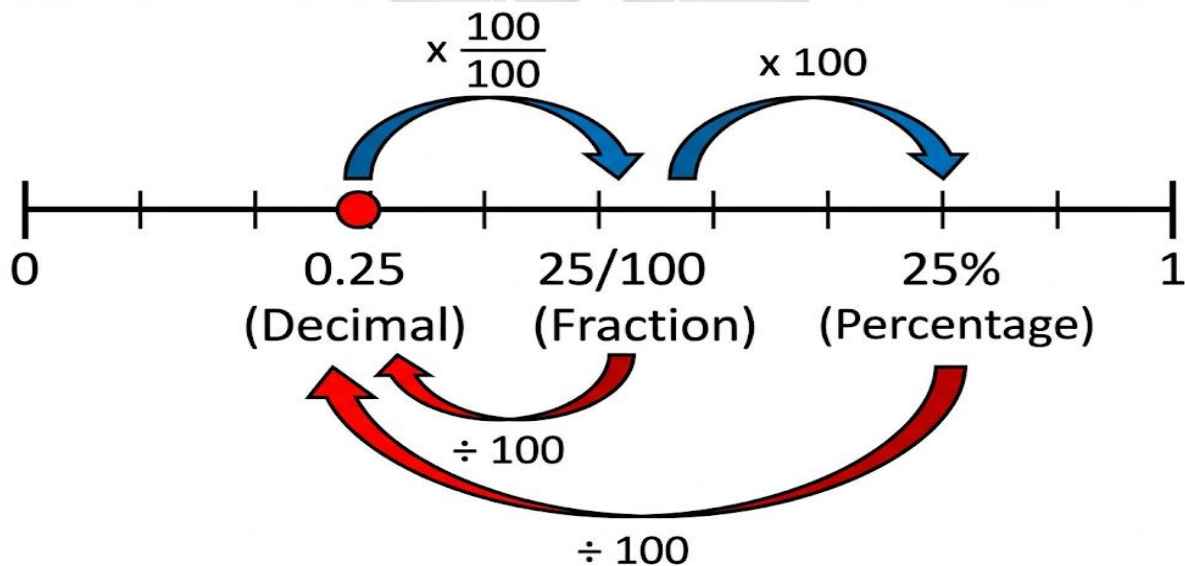
To change any fraction to a percentage, make its denominator 100. How? Multiply the fraction by 100%.

- **Step 1:** Multiply the fraction by 100.
- **Step 2:** Do the multiplication and write the % sign.
- **Example:** Convert $\frac{3}{4}$ to a percentage.
 - $\frac{3}{4} \times 100 = (3 \times 100) / 4 = 300 / 4 = 75$. So, $\frac{3}{4} = 75\%$.
- **Example:** Convert $\frac{2}{5}$ to a percentage.
 - $\frac{2}{5} \times 100 = (2 \times 100) / 5 = 200 / 5 = 40\%$.

3. Converting Decimals to Percentages

A decimal is also a fraction (e.g., $0.5 = 5/10$). To convert a decimal to a percentage, multiply it by 100, which means move the decimal point two places to the **right**.

- **Example:** Convert 0.45 to a percentage.
 - $0.45 \times 100 = 45\%$. (The decimal point moves from 0.45 to 45.)
- **Example:** Convert 0.6 to a percentage.
 - $0.6 \times 100 = 60\%$. (We write 0.6 as 0.60, then move the point: 060. → 60%)



4. Converting Percentages to Fractions and Decimals

- **Percentage to Fraction:** Remember, % means "per hundred". So, write the percentage as a fraction over 100, then simplify.
 - **Example:** $20\% = 20/100$. Simplify by dividing numerator and denominator by 20: $20 \div 20 = 1$, $100 \div 20 = 5$. So, $20\% = \frac{1}{5}$.
- **Percentage to Decimal:** Since % means $\div 100$, to convert a percentage to a decimal, divide by 100, which means move the decimal point two places to the **left**.
 - **Example:** $35\% = 35.0\% \rightarrow$ Move decimal two places left: **0.35**.
 - **Example:** $5\% = 05.0\% \rightarrow$ Move decimal two places left: **0.05**.

5. Finding a Percentage of a Quantity

The word "**of**" in mathematics often means "**multiply**". To find a percentage of any number:

1. Change the percentage to a fraction or decimal.
 2. Multiply that fraction or decimal by the number.
- **Example:** Find 10% of 200 oranges.
 - Method 1 (Fraction): $10\% = 10/100 = 1/10$. $(1/10) \times 200 = 200/10 = \mathbf{20 \text{ oranges}}$.
 - Method 2 (Decimal): $10\% = 0.10$. $0.10 \times 200 = \mathbf{20 \text{ oranges}}$.
 - **Real-life link:** If you save 20% of your N500 weekly allowance, how much do you save?
 $20\% \text{ of N}500 = 0.20 \times 500 = \mathbf{N100}$.

EVALUATION

1. What does "percent" mean? What is its symbol?
2. Convert $\frac{1}{2}$ to a percentage. Explain how it shows "half".
3. Change 0.08 to a percentage.
4. Express 40% as a fraction in its simplest form.
5. Change 8% to a decimal.
6. Calculate 25% of 800 Naira. Show two methods.
7. In a basket of 60 fruits, 15% are bad. How many fruits are good?

ASSIGNMENT

1. **(Oral/Written)** Ask a family member what "50% off" means in a market. Explain it to them using the idea of "out of 100".
 2. **(Drawing)** Draw a rectangle and shade 65% of it. Below your drawing, write the shaded part as a fraction and a decimal.
 3. **(Written)** A farmer has 80 chickens. If 30% are roosters, how many are hens? (Hint: Find 30% first, then subtract from the total).
-



WEEK 2: PERCENTAGES II (APPLICATIONS)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Calculate percentage increase and decrease.
2. Determine profit and loss percentages.
3. Explain the meaning of Simple Interest and solve simple problems.
4. Solve word problems involving VAT (Value Added Tax) and discounts.

CONTENT

1. Percentage Increase and Decrease

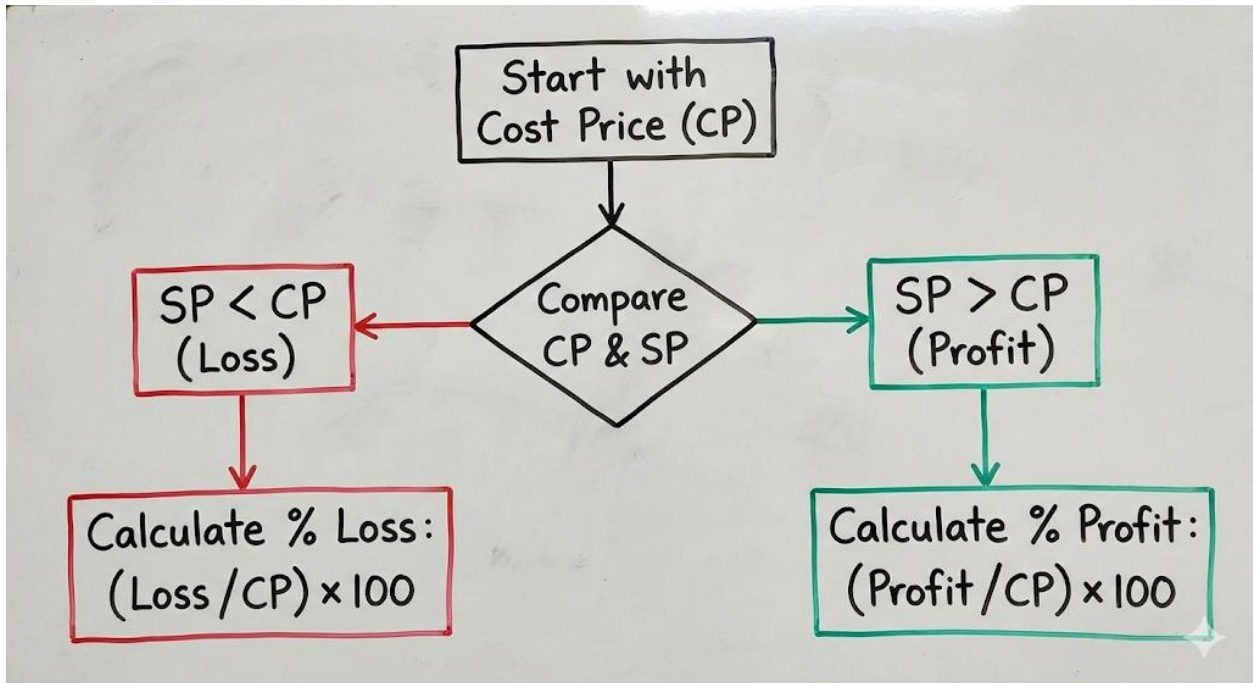
We calculate this to know how much a price, population, or quantity has gone UP (increase) or DOWN (decrease).

- **Step 1:** Find the **Actual Change**. New Value - Old Value. (If new is smaller, the change is a decrease).
- **Step 2:** Use the formula: **$(\text{Change} \div \text{Original Value}) \times 100\%$** .
- **Example (Increase):** A textbook price rose from ₦500 to ₦600.
 - $\text{Change} = 600 - 500 = \text{₦}100$.
 - $\% \text{ Increase} = (100 \div 500) \times 100\% = 0.2 \times 100\% = \mathbf{20\%}$.
- **Example (Decrease):** The price of rice fell from ₦40,000 per bag to ₦36,000.
 - $\text{Change} = 36,000 - 40,000 = \text{₦}4,000$ (a decrease).
 - $\% \text{ Decrease} = (4,000 \div 40,000) \times 100\% = 0.1 \times 100\% = \mathbf{10\%}$.

2. Profit and Loss Percent

This is very important for business (like your parents in the market).

- **Cost Price (CP):** Price you buy an item.
- **Selling Price (SP):** Price you sell an item.
- **Profit** = $\text{SP} - \text{CP}$ (when SP is greater). **Loss** = $\text{CP} - \text{SP}$ (when CP is greater).
- **Percentage Profit/Loss** = **$(\text{Profit or Loss} \div \text{CP}) \times 100\%$** .



Example: Auntie Ngozi bought a wrapper for ₦2,000 and sold it for ₦2,500.

- Profit = ₦2,500 - ₦2,000 = ₦500.
- % Profit = $(500 \div 2000) \times 100\% = 0.25 \times 100\% = 25\%$.

3. Discount and VAT

- **Discount:** A reduction in price to attract buyers. You **subtract** the discount from the original price.
 - **Example:** A shoe priced ₦5,000 has a 20% discount.
 - Discount Amount = 20% of ₦5,000 = ₦1,000.
 - Price to Pay = ₦5,000 - ₦1,000 = ₦4,000.
- **Value Added Tax (VAT):** A small government charge **added** to the price of goods and services. You **add** VAT to the original price.
 - **Example:** A bill for food is ₦2,000. VAT is 7.5%.
 - VAT Amount = 7.5% of ₦2,000 = ₦150.
 - Total to Pay = ₦2,000 + ₦150 = ₦2,150.

4. Simple Interest

This is money earned on savings or paid on a loan. It is called "simple" because it is calculated only on the original amount (Principal) every year.

- **Formula: Interest (I) = (Principal × Rate × Time) / 100**
- **Principal (P):** The original amount of money saved or borrowed.
- **Rate (R):** The percentage interest per year.
- **Time (T):** The number of years the money is saved or borrowed.
- **Example:** If you save ₦10,000 in a bank for 2 years at 5% per annum:
 - $I = (10,000 \times 5 \times 2) / 100 = (100,000) / 100 = \text{₦}1,000.$
 - After 2 years, you will have $\text{₦}10,000 + \text{₦}1,000 = \text{₦}11,000.$

EVALUATION

1. The population of a village increased from 1,200 to 1,500. What is the percentage increase?
2. A trader bought goods for ₦800 and sold them for ₦720. Calculate her percentage loss.
3. Write down the Simple Interest formula. What does 'T' represent?
4. Find the simple interest on ₦5,000 for 3 years at 10% per annum.
5. Calculate the VAT (at 7.5%) on a new uniform that costs ₦3,000.
6. A phone is on sale with a 15% discount. If its original price is ₦40,000, what is the sale price?

ASSIGNMENT

1. **(Practical)** Interview a petty trader at home or nearby. Ask for the cost price and selling price of one item. Calculate their profit or loss percentage.
2. **(Written)** You borrow ₦15,000 from a cooperative to buy books. They charge 8% simple interest per year. How much interest will you pay after 2 years? How much will you pay back in total?
3. **(Oral/Written)** Explain the difference between a discount and VAT to a younger sibling using a toy as an example.

WEEK 3: DEMOGRAPHY & MONEY I

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define population and census.
2. Distinguish between birth rate and death rate.
3. Explain population density.
4. List two economic consequences of overpopulation.

CONTENT

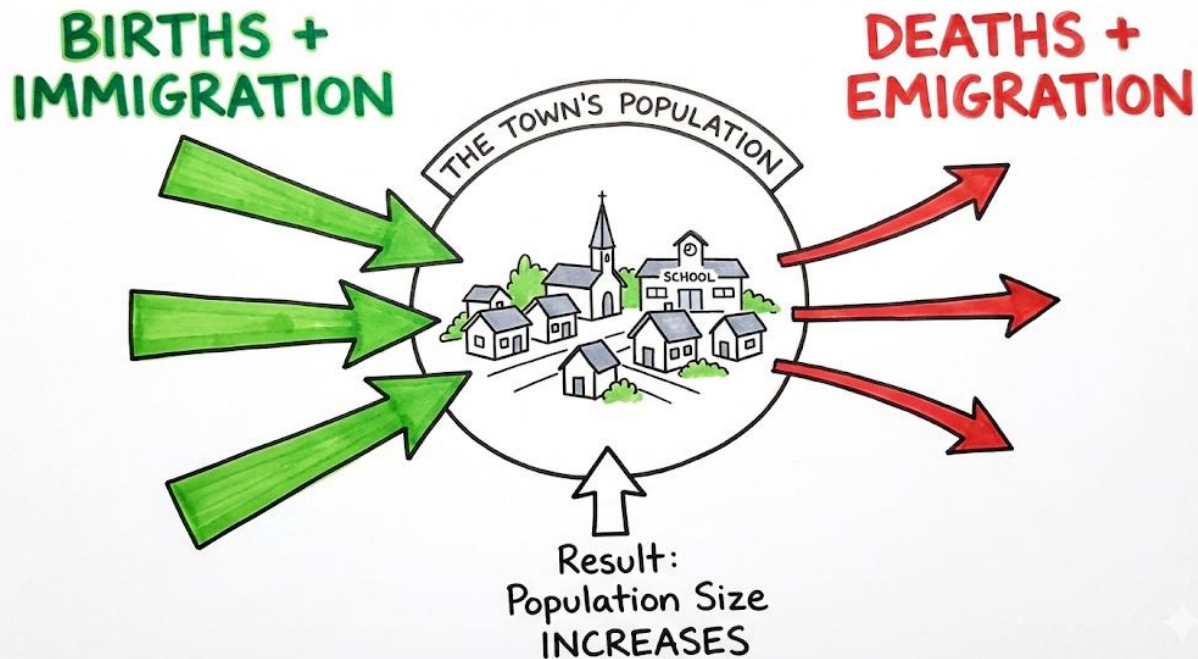
1. Meaning of Population and Census

- **Population:** The total number of people living in a particular place at a particular time. This place can be a compound, a village, a state (like Lagos State), or a country (like Nigeria).
- **Census:** The official counting of all the people and their important details (like age, job) in a country. In Nigeria, the National Population Commission (NPC) tries to do this every 10 years. The last one was in 2006. A census helps the government plan for schools, hospitals, and roads.

2. How Population Changes: Birth Rate, Death Rate, Migration

A population can grow, shrink, or stay the same because of three things:

- **Birth Rate:** This is the number of live babies born in a year for every 1,000 people. A high birth rate makes the population grow. Think of a family with many children.
- **Death Rate:** This is the number of people who die in a year for every 1,000 people. A high death rate (from sickness, accidents) makes the population smaller.
- **Migration:** This is the movement of people into or out of a place.
 - **Immigration:** People moving *into* a place (e.g., people moving from village to Lagos for work). This *increases* population.
 - **Emigration:** People moving *out* of a place (e.g., people leaving Nigeria for "Japa"). This *decreases* population.



3. Population Density

This tells us if a place is crowded or has plenty of space. It is calculated as: **Population Density** = **Total Population** ÷ **Total Land Area**.

- **Example:** If 10,000 people live in a village that is 50 square kilometres large.
 - Density = $10,000 \div 50 = 200$ people per square kilometre.
- **High Density (Crowded):** Lagos Island, Mushin. Lots of people in a small space.
- **Low Density (Plenty of Space):** Some parts of Niger State or the Sahara. Few people in a large area.

4. Consequences of Population Size

- **Overpopulation (Too many people for the available resources):**
 - **Shortage:** Not enough food, water, houses, or jobs for everyone.
 - **Pressure:** Overcrowded schools and hospitals, traffic jams, long queues.
 - **Pollution:** More waste and dirt in the environment.
- **Underpopulation (Too few people for the available land and resources):**
 - **Waste:** Land and natural resources (like minerals, forests) are not used.
 - **Shortage of Labour:** Not enough people to work in farms, factories, or hospitals.

- **Less Development:** Fewer people to build businesses and develop the area.

EVALUATION

1. What is the main purpose of conducting a national census?
2. If a town has more births and immigrants than deaths and emigrants, what will happen to its population?
3. Define "Population Density" in your own words.
4. Why is Lagos State more crowded (has higher density) than Taraba State?
5. Mention two problems your community might face if the population suddenly doubled.
6. What is the difference between immigration and emigration?

ASSIGNMENT

1. **(Oral/Research)** Ask your parents or guardian: "What was the population of our Local Government Area (LGA) in the last census?" Try to find out.
 2. **(Written/Calculation)** If a local government area has 200,000 people living on 400 km² of land, calculate its population density. What does this number tell you?
 3. **(Practical/Drawing)** Draw two pictures side-by-side. Picture A: Show an overpopulated city with traffic, queues. Picture B: Show an underpopulated village with plenty of empty land. Write one sentence under each picture.
-

WEEK 4: MONEY II - CURRENCY CONVERSION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify the Nigerian currency and major foreign currencies (Dollar, Pound, Euro).
2. Explain the concept of exchange rate.
3. Convert Naira to foreign currencies and vice versa using given rates.
4. Solve word problems related to travel money.

CONTENT

1. Nigerian and Foreign Currencies

Every country has its own currency (money). We must know them, especially if our family travels or does business abroad.

- **Nigeria:** Naira (₦) and Kobo (k). 100 kobo = 1 Naira. (e.g., ₦500, ₦1,000 notes).
- **United States of America (USA):** **Dollar (\$) **. (e.g., 1, 5, 20 bills).
- **United Kingdom (UK/Britain):** Pound Sterling (£). (e.g., £5, £10, £20 notes).
- **European Union** (Countries like Germany, France): Euro (€).
- **Ghana:** Cedi (GH¢).
- **West African CFA Zone** (Benin, Niger etc.): CFA Franc (CFA).



Nigerian Naira (₦)



US Dollar (\$)



British Pound (£)



Euro (€)

2. Understanding Exchange Rate

You cannot use Naira to buy things in America. You need Dollars. The **Exchange Rate** is the price of one country's currency in terms of another. It changes daily.

- **Example Rate (for teaching):** \$1 (1 US Dollar) = ₦1,000 (One thousand Naira).
- This means to buy 1 Dollar, you need to pay ₦1,000. The Naira is "weaker" because you need many Naira to buy 1 Dollar.

3. How to Convert

- **Converting FOREIGN to NAIRA (Multiply):** If you have foreign currency and want to know its value in Naira, **MULTIPLY** by the exchange rate.
 - **Example:** How much is ****5**** in Naira if $1 = ₦1000$?
 - $5 \times ₦1000 = ₦5,000$.
- **Converting NAIRA to FOREIGN (Divide):** If you have Naira and want to buy foreign currency, **DIVIDE** by the exchange rate.
 - **Example:** How many Dollars can you get with ₦8,000 if $1 = ₦1000$?
 - $₦8,000 \div ₦1000 = 8$.

4. Real-Life Application: Bureau de Change and Travel

Places where you change money are called Bureau de Change or banks.

- **Scenario:** Your uncle is traveling to the UK. He needs Pounds (£). He goes to a Bureau de Change with ₦120,000. If the rate is $£1 = ₦1,500$, how many Pounds will he get?
 - $₦120,000 \div ₦1,500/£ = £80$.
- When he returns with £20 left, he sells it back at the same rate: $£20 \times ₦1,500 = ₦30,000$.

EVALUATION

1. What is the currency of the United Kingdom and its symbol?
2. What does an exchange rate of $£1 = ₦1,800$ mean?
3. If $€1 = ₦1,200$, convert €10 to Naira.
4. Using the same rate ($€1 = ₦1,200$), convert ₦6,000 to Euros.

5. Why do people need to change money when they travel to another country?
6. A student in Ghana needs to pay a ₦5,000 Nigerian exam fee. If GH¢1 = ₦100, how many Cedis does she need?

ASSIGNMENT

1. **(Drawing)** Create a "Currency Passport". Draw and label the symbols for Naira, Dollar, Pound, and Euro. Write one country that uses each.
2. **(Written)** If the exchange rate at the airport is $1 = ₦950$, calculate:
a) The cost in Naira of a \$50 video game.
b) How many Dollars you can get with ₦19,000.
3. **(Oral/Practical)** Ask a family member or neighbor who has traveled abroad: "Which foreign currency did you use? What was the exchange rate like?" Share the answer in class.



WEEK 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7: ALGEBRAIC PROCESSES - OPEN SENTENCES

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a variable (unknown).
2. Solve open sentences involving addition and subtraction.
3. Solve open sentences involving multiplication and division.
4. Form simple equations from word problems.

CONTENT

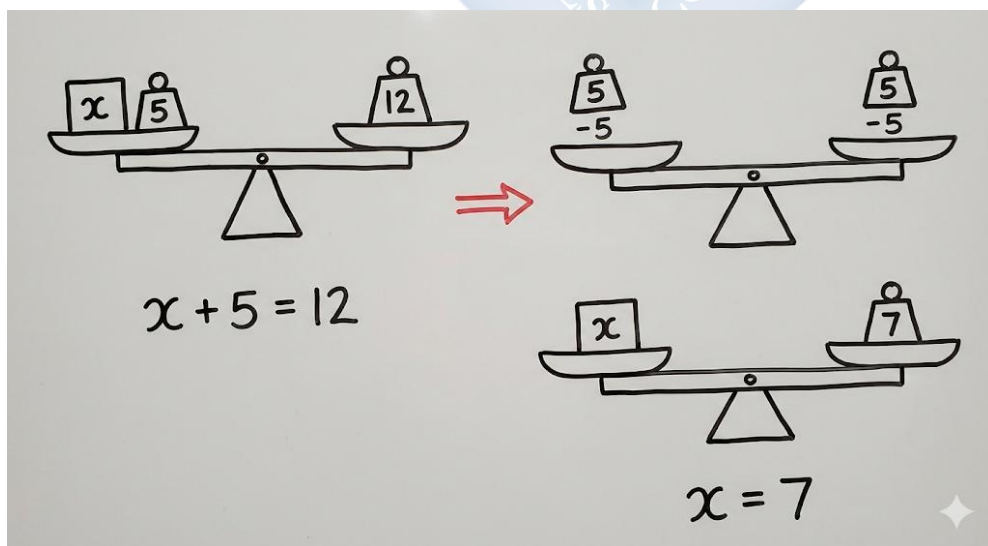
1. Understanding Variables (The Unknown)

In mathematics, sometimes we don't know a number yet. We use a letter (like **x**, **y**, **a**, **n**) or a box ($\square$) to hold the place of that unknown number. This is called a **variable**.

- **Example:** $\square + 5 = 12$. The box is the unknown. We can think: "What number plus 5 gives 12?" The answer is 7.
- We can use letters: $n + 5 = 12$. Here, **n** is the variable.

2. Solving Open Sentences: The Balance Method

Think of an equation like a balanced scale. Whatever you do to one side, you must do to the other to keep it balanced. To find the unknown, we perform the **opposite (inverse)** operation.



- **For Addition (+), the opposite is Subtraction (-).**
 - $x + 8 = 15$. To get x alone, remove $+8$ by subtracting 8 from **both sides**.
 - $x + 8 - 8 = 15 - 8$.
 - So, $x = 7$.
- **For Subtraction (-), the opposite is Addition (+).**
 - $y - 4 = 10$. To get y alone, remove -4 by adding 4 to **both sides**.
 - $y - 4 + 4 = 10 + 4$.
 - So, $y = 14$.

3. Solving for Multiplication ($\times$) and Division ($\div$)

- **For Multiplication ($\times$), the opposite is Division ($\div$).**
 - $3 \times p = 18$ (or $3p = 18$). Here, 3 is multiplied by p . The opposite is to divide both sides by 3.
 - $3p \div 3 = 18 \div 3$.
 - So, $p = 6$.
- **For Division ($\div$), the opposite is Multiplication ($\times$).**
 - $k \div 5 = 4$ (or $k/5 = 4$). Here, k is divided by 5. The opposite is to multiply both sides by 5.
 - $(k/5) \times 5 = 4 \times 5$.
 - So, $k = 20$.

4. Forming Equations from Word Problems

Read the problem carefully. Look for the unknown and what is happening to it.

- **Problem:** "I think of a number, multiply it by 2, and get 16. What is the number?"
 - Let the number be **m** .
 - Multiply by 2: $2 \times m$ or **$2m$** .
 - It gives 16: **$2m = 16$** .
 - Solve: $m = 16 \div 2 = 8$.

EVALUATION

1. If $a + 12 = 30$, what is the value of a ? Show your balancing step.
2. Solve for b : $b - 7 = 15$.
3. Find y if $4y = 28$.
4. Find x if $x \div 6 = 5$.
5. Write an equation for: "A number divided by 4 gives 3." Then solve it.
6. I had some sweets. I gave away 9 and have 11 left. How many did I have at first? (Form an equation and solve).

ASSIGNMENT

1. **(Written)** Solve for the unknown: a) $25 + m = 40$ b) $n / 8 = 4$
2. **(Practical)** Write two "I think of a number" puzzles for your friend. One must involve addition/subtraction, the other multiplication/division. Solve each other's puzzles.
3. **(Oral/Written)** Explain to a classmate, using simple words, what the "balance method" means in algebra.

WEEK 8: MEASUREMENT - LENGTH, WEIGHT & TIME I

OBJECTIVES By the end of the lesson, pupils should be able to:

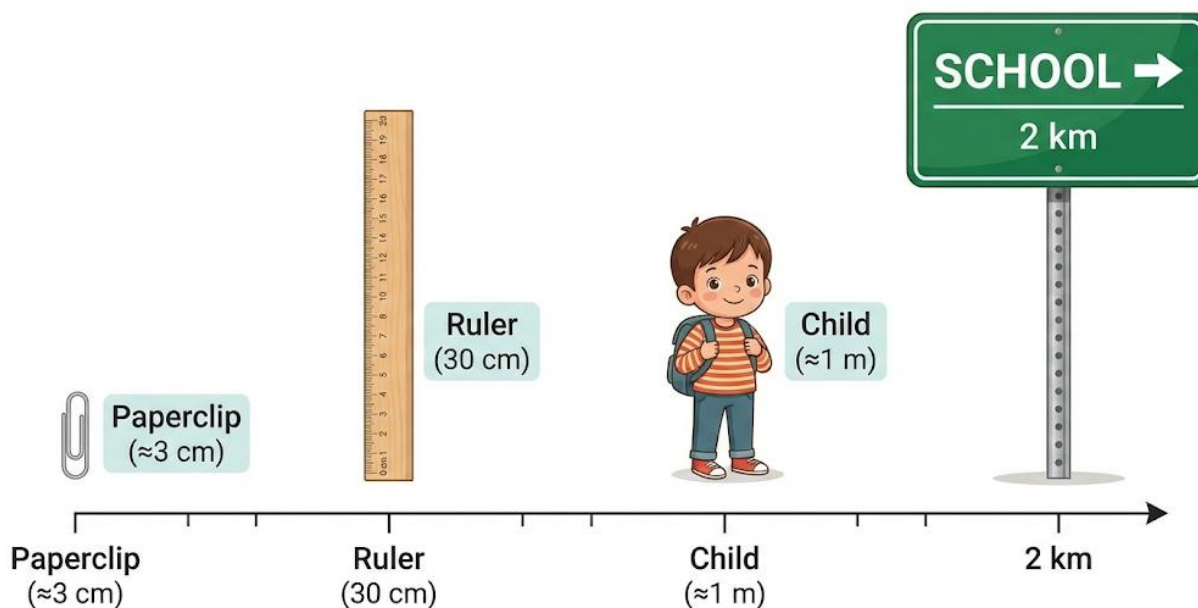
1. List the metric units for length, weight, and time.
2. Convert metres to kilometres and centimetres to metres.
3. Convert grams to kilograms.
4. Estimate distances and weights of common objects.

CONTENT

1. Units of Length

Length measures how long, tall, wide, or far something is. We use the metric system.

- **Millimetre (mm):** Very tiny. Thickness of a 10 kobo coin or a key.
- **Centimetre (cm):** Small. Width of your fingernail or a pencil. **10 mm = 1 cm.**
- **Metre (m):** Medium. The height of a door, or your arm span. **100 cm = 1 m.**
- **Kilometre (km):** Long distance. From your school to the main market, or between cities. **1,000 m = 1 km.**



- **To convert:**

- m to km: Divide by 1,000. (e.g., 2,500 m = $2,500 \div 1,000 = 2.5$ km).
- km to m: Multiply by 1,000. (e.g., 0.75 km = $0.75 \times 1,000 = 750$ m).
- cm to m: Divide by 100. (e.g., 250 cm = $250 \div 100 = 2.5$ m).

2. Units of Weight (Mass)

Weight measures how heavy something is. We often say "mass".

- **Gram (g):** Very light. A sachet of sugar (≈ 10 g), a teaspoon of salt.
- **Kilogram (kg):** Heavy. A bag of rice (50kg), a big textbook. **1,000 g = 1 kg.**
- **Tonne (t):** Very heavy. A car, a small truck. 1,000 kg = 1 tonne.
- **To convert g to kg:** Divide by 1,000. (e.g., 3,500 g = $3,500 \div 1,000 = 3.5$ kg).

3. Units of Time

Time measures how long an event lasts.

- 60 seconds (sec) = 1 minute (min)
- 60 minutes = 1 hour (hr)
- 24 hours = 1 day
- 7 days = 1 week
- About 4 weeks = 1 month
- 365 days = 1 year (366 in a leap year)

4. Estimation

Making a good guess using what you know.

- **Length:** The length of your classroom is about **10 metres**. The distance from your house to school might be about **2 kilometres**.
- **Weight:** A bottle of 50cl water weighs about **500 grams**. A newborn baby weighs about **3 kilograms**.

EVALUATION

1. Which unit (mm, cm, m, km) would you use to measure: a) The length of your desk? b) The distance from Lagos to Abuja?
2. Convert 5 kilometres into metres.
3. A bag of flour weighs 2,500 grams. What is its weight in kilograms?
4. How many centimetres are there in 3.5 metres?
5. How many minutes are there in 3.5 hours?
6. Estimate: Is the weight of your mathematics textbook closer to 2 grams, 2 kilograms, or 2 tonnes?

ASSIGNMENT

1. **(Practical)** Use a tape measure or ruler to measure and record in cm: a) Your maths textbook length. b) Your teacher's table width. Convert both measurements to metres.
2. **(Written)** Convert the following: a) 4,200 m = ____ km b) 8.2 kg = ____ g
3. **(Oral/Estimation)** At home, find three items (e.g., a tin of milk, a bucket, a pillow). Estimate their weight in grams or kg, then check with an adult if possible. Share your estimates in class.

WEEK 9: MEASUREMENT - TIME & SPEED

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Convert time between 12-hour and 24-hour clock systems.
2. State the formula for speed.
3. Calculate speed given distance and time.
4. Calculate distance given speed and time.

CONTENT

1. 12-Hour vs. 24-Hour Clock

- **12-Hour Clock:** This is what we use in everyday conversation. It divides the day into two 12-hour periods: **a.m.** (ante meridiem - before noon, from midnight to 11:59 midday) and **p.m.** (post meridiem - after noon, from 12:00 noon to 11:59 night).
 - Example: 7:00 a.m. (morning), 2:30 p.m. (afternoon), 9:00 p.m. (night).
- **24-Hour Clock (Military/International Time):** Used by the military, airlines, and for clarity. It runs from 00:00 (midnight) to 23:59 (one minute to midnight). There is no a.m. or p.m.
 - **Rule to convert PM to 24-hour:** Add 12 to the hour.
 - **Rule to convert AM to 24-hour:** Keep the hour, but write 0 before single digits.
 - **Examples:**
 - 3:15 p.m. → $3 + 12 = 15:15$.
 - 8:00 a.m. → **08:00**.
 - 12:00 noon → **12:00**.
 - 12:00 midnight → **00:00**.

2. Understanding Speed

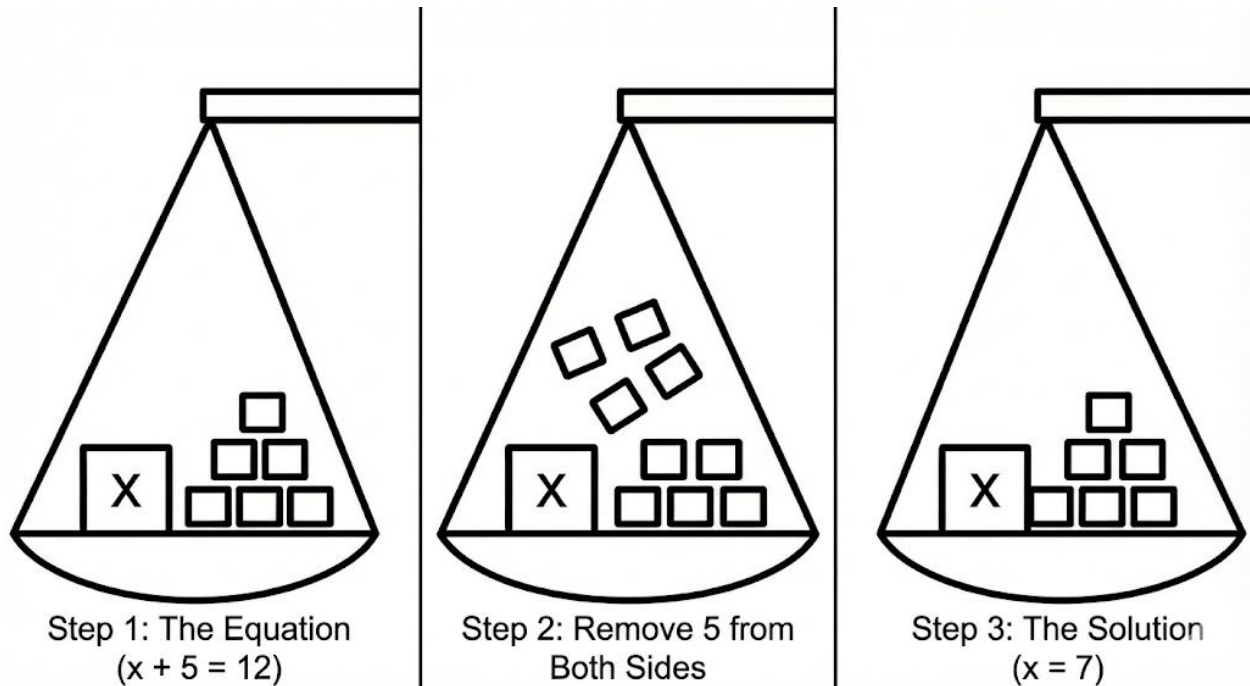
Speed tells us how fast something is moving. It is the distance covered in a unit of time.

- **Formula:** $\text{Speed} = \text{Distance} \div \text{Time}$.
- **Units:** Kilometres per hour (km/h), Metres per second (m/s). "**Per**" means "for every".

- If a car's speed is 60 km/h, it means the car can travel 60 kilometres in 1 hour if it keeps that speed.

3. The Speed-Distance-Time Triangle

This is a magic tool to remember all three formulas!



- To find **Speed (S)**: Cover S with your finger. You see D over T. So, $S = D / T$.
- To find **Distance (D)**: Cover D. You see S next to T. So, $D = S \times T$.
- To find **Time (T)**: Cover T. You see D over S. So, $T = D / S$.

4. Solving Problems

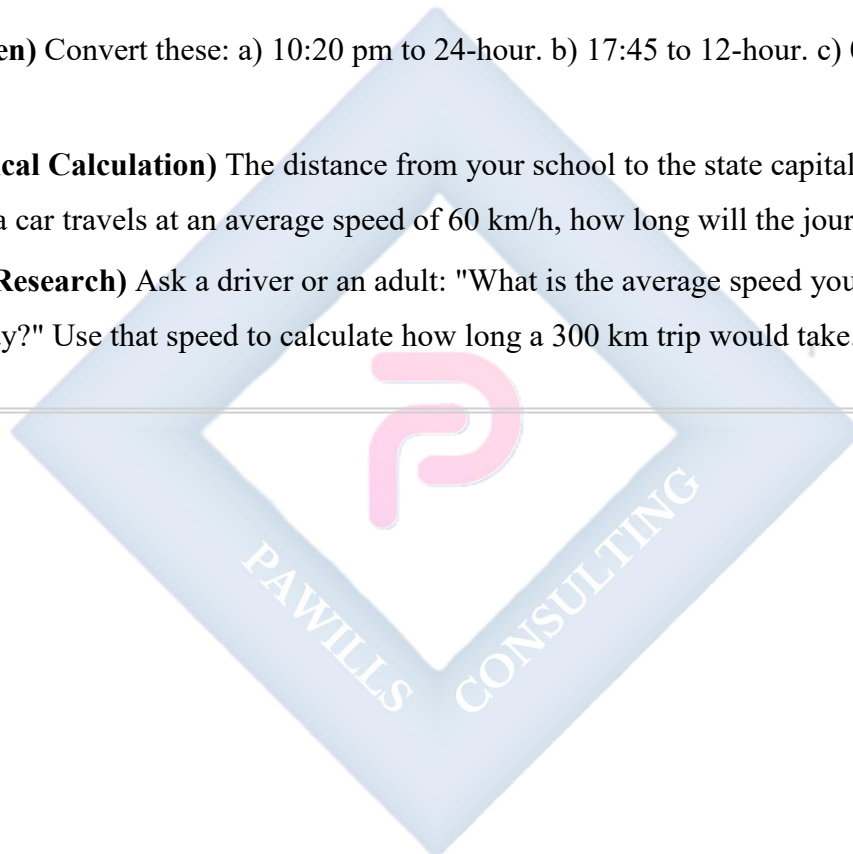
- **Finding Speed**: A bus travels 120 km in 2 hours. Speed = $120 \text{ km} \div 2 \text{ hr} = \mathbf{60 \text{ km/h}}$.
- **Finding Distance**: If you walk at 4 km/h for 3 hours, how far? Distance = $4 \text{ km/h} \times 3 \text{ h} = \mathbf{12 \text{ km}}$.
- **Finding Time**: How long does it take to fly 800 km at a speed of 400 km/h? Time = $800 \text{ km} \div 400 \text{ km/h} = \mathbf{2 \text{ hours}}$.

EVALUATION

1. Convert 5:45 p.m. to 24-hour time.
2. What is 14:30 in 12-hour time?
3. Write the formula for calculating speed.
4. A cyclist covers 30 km in 1.5 hours. What is his average speed?
5. If a plane flies at 500 km/h, how far will it travel in 4 hours?
6. How long will it take to drive 240 km at a speed of 80 km/h? (Use the triangle).

ASSIGNMENT

1. **(Written)** Convert these: a) 10:20 pm to 24-hour. b) 17:45 to 12-hour. c) 00:20 to 12-hour.
2. **(Practical Calculation)** The distance from your school to the state capital is roughly 120 km. If a car travels at an average speed of 60 km/h, how long will the journey take?
3. **(Oral/Research)** Ask a driver or an adult: "What is the average speed you drive on the highway?" Use that speed to calculate how long a 300 km trip would take.



WEEK 10: MEASUREMENT - PERIMETER, AREA & VOLUME

OBJECTIVES By the end of the lesson, pupils should be able to:

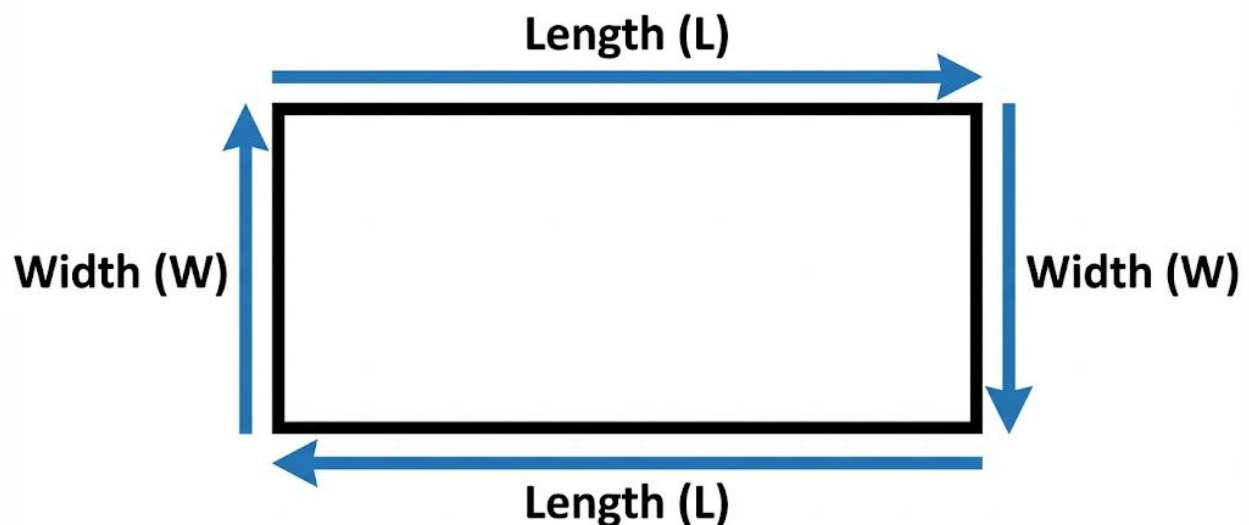
1. Calculate the perimeter of regular and irregular shapes.
2. Calculate the area of rectangles, squares, and triangles.
3. Calculate the volume of cubes and cuboids.
4. Convert between litres and millilitres.

CONTENT

1. Perimeter

Perimeter is the total distance around the outside edge of a flat (2D) shape. Imagine walking around the boundary of a football field; the distance you walk is the perimeter.

- **For any shape:** Add up the lengths of all the sides.
- **Rectangle:** Perimeter = $2 \times (\text{Length} + \text{Width})$. (Because it has two lengths and two widths).
- **Square:** Perimeter = $4 \times \text{Side Length}$.

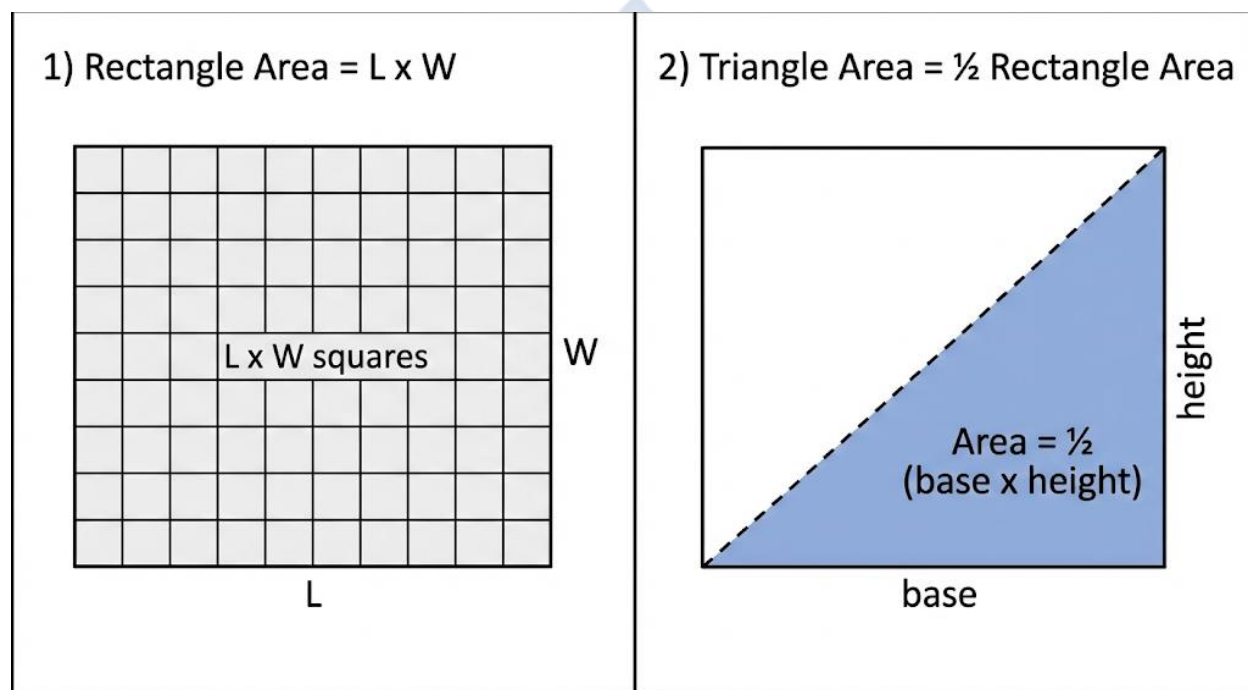


$$P = L + W + L + W = 2L + 2W$$

2. Area

Area is the amount of surface a shape covers. It is measured in square units (cm^2 , m^2). Think of it as the number of square tiles needed to cover the floor of a room.

- **Rectangle:** Area = Length $\times$ Width. (e.g., A room 5m long and 3m wide has an area of 15 square metres (15m^2)).
- **Square:** Area = Side $\times$ Side = Side².
- **Triangle:** Area = $\frac{1}{2} \times$ Base $\times$ Height. The height must be the **perpendicular height** (a straight line from the base to the top corner, forming a right angle).



3. Volume

Volume is the amount of space inside a 3-dimensional (3D) solid object. It tells us how much the object can hold. It is measured in cubic units (cm^3 , m^3).

- **Cube or Cuboid (Box):** Volume = Length $\times$ Width $\times$ Height.
- **Example:** A cartoon milk box 10 cm long, 5 cm wide, and 20 cm high.
 - Volume = $10 \times 5 \times 20 = 1,000 \text{ cm}^3$.

4. Capacity

Capacity is a special type of volume—it measures how much **liquid** a container can hold. The main units are millilitres (ml) and Litres (L).

- **1,000 millilitres (ml) = 1 Litre (L).**
- **Important Link:** A cube with sides of 10 cm (i.e., 10 cm x 10 cm x 10 cm = 1,000 cm³) has a volume that can hold exactly 1 Litre of water.
- **To convert L to ml:** Multiply by 1,000. (e.g., 2.5 L = $2.5 \times 1,000 = 2,500$ ml).

EVALUATION

1. Find the perimeter of a square whose side is 7 cm.
2. A rectangular farm is 50 m long and 30 m wide. Calculate its area.
3. A triangular flag has a base of 12 cm and a height of 8 cm. What is its area?
4. Calculate the volume of a box with dimensions: Length=6cm, Width=4cm, Height=3cm.
5. How many millilitres are there in 4.2 Litres?
6. A square has a perimeter of 36 cm. What is the length of one side? What is its area?

ASSIGNMENT

1. **(Drawing/Calculation)** Draw a rectangle of length 9 cm and width 6 cm. Calculate and write its Perimeter and Area inside the drawing.
 2. **(Practical)** Find a rectangular box at home (e.g., cereal box, soap box). Measure its length, width, and height in cm. Calculate its volume in cm³.
 3. **(Oral/Written)** Look at a 1.5L bottle of soft drink or water. How many millilitres is that? If you have a 200ml cup, how many full cups can you get from the bottle?
-

PRIMARY 6 MATHEMATICS SCHEME OF WORK

THIRD TERM

Week	Topic	Content
1	Plane Figures and Properties	Classification of plane shapes - triangles (equilateral, isosceles, scalene), quadrilaterals (square, rectangle, parallelogram, rhombus, trapezium); Properties of shapes; Symmetry - lines of symmetry; Congruence and similarity
2	Angles	Types of angles - acute, right, obtuse, straight, reflex; Measuring angles with protractor; Drawing angles; Angles on a straight line; Angles at a point; Vertically opposite angles; Angles in triangles and quadrilaterals
3	Height and Distances	Measuring heights using indirect methods; Scale and proportion in measurement; Using similar triangles; Practical applications; Estimation techniques; Map reading and distance calculation
4	Polygons and Scale Drawing	Types of polygons - pentagon, hexagon, heptagon, octagon, etc.; Regular and irregular polygons; Properties of polygons; Interior and exterior angles; Scale drawing - meaning and uses; Drawing to scale; Reading scale drawings; Maps and plans
5	MIDTERM EXAMINATION	Assessment of topics covered in weeks 1-4
6	MIDTERM BREAK	Rest and relaxation period
7	Everyday Statistics - Population	Collecting and organizing population data; Population tables and graphs; Population pyramid; Birth and death statistics; Migration patterns; Population projection; Interpreting population data; Demographics and planning
8	Measures of Central Tendency I	Mean (average) - calculating and interpreting; Finding mean from data sets; Mean from frequency tables; Practical applications of mean; Weighted mean introduction

9	Measures of Central Tendency II	Median - finding the middle value; Median from ordered data; Mode - finding the most frequent value; Bimodal data; Range - measure of spread; Comparing mean, median, and mode; Choosing appropriate measures; Real-life applications
10	Comprehensive Problem Solving & Secondary School Preparation	Mixed problems combining multiple concepts; Multi-step word problems; Real-life mathematical applications; Problem-solving strategies; Critical thinking in mathematics; Calculator skills; Mathematics in secondary school - topics preview; Study skills and examination techniques
11	REVISION	Comprehensive review of Primary 6 topics; Integration of concepts; Practice examinations; Problem-solving practice; Secondary school entrance exam preparation; Question and answer sessions
12	EXAMINATION	End of term and final primary school assessment
13	CLOSING	Graduation ceremony; Report cards; Secondary school preparation; Celebration of achievements

PRIMARY 6 MATHEMATICS LESSON NOTES

THIRD TERM

WEEK 1: PLANE FIGURES AND PROPERTIES

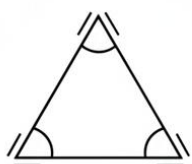
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Identify and describe different types of triangles based on their sides.
2. Distinguish between different types of quadrilaterals (Square, Rectangle, Parallelogram, Rhombus, Trapezium).
3. Identify lines of symmetry in given shapes.
4. Explain the difference between congruent and similar shapes.

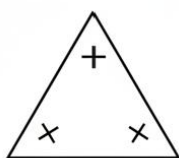
CONTENT

1. Types of Triangles

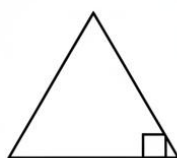
A triangle is a flat, closed shape with three straight sides and three angles. We can name and group triangles by looking at the length of their sides.



Isosceles Triangle



Equilateral Triangle



Right-Angled Triangle

- **Equilateral Triangle:**

- All three sides are equal in length.
- Because the sides are equal, all three angles are also equal. Each angle measures 60° .
- Real-life example: The yield sign on some roads, or the sides of a triangular set-square.

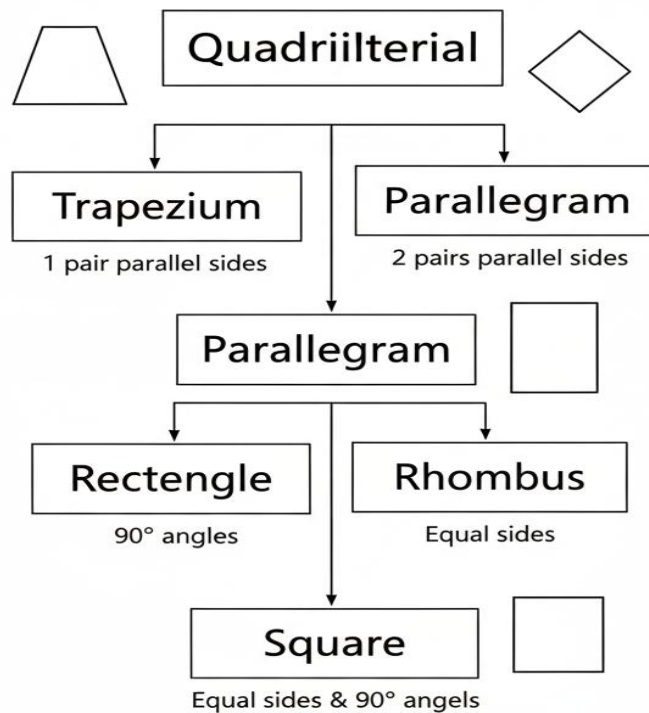
- **Isosceles Triangle:**

- Only **two** sides are equal in length. The third side is different.

- The angles opposite the two equal sides are also equal. It looks balanced.
- Real-life example: The roof of many houses, a slice of pizza.
- **Scalene Triangle:**
 - **No** sides are equal. All three sides have different lengths.
 - All three angles are also different.
 - Real-life example: A randomly broken piece of a slate, a triangular scarf with sides of different lengths.

2. Types of Quadrilaterals

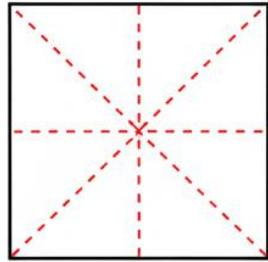
A quadrilateral is any flat, closed shape with four straight sides. There are special types we must know.



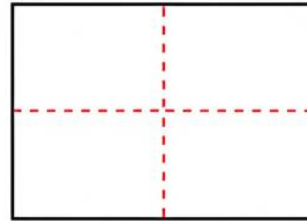
- **Square:**
 - **Properties:** All four sides are equal. All four angles are right angles (90°). Opposite sides are parallel.
 - Real-life example: A chessboard square, a window pane.
- **Rectangle:**
 - **Properties:** Opposite sides are equal and parallel. All four angles are right angles (90°).
 - Real-life example: A door, a sheet of paper, a mobile phone screen.
- **Parallelogram:**
 - **Properties:** Opposite sides are equal and parallel. Opposite angles are equal. It looks like a rectangle that has been pushed or slanted.
 - Real-life example: The design on some school bags, the face of a ramp.
- **Rhombus:**
 - **Properties:** All four sides are equal. Opposite sides are parallel. Opposite angles are equal. It is like a pushed square.
 - Real-life example: A kite shape, the diamonds on some playing cards.
- **Trapezium (US: Trapezoid):**
 - **Properties:** It has **only one** pair of parallel sides. The other two sides are not parallel.
 - Real-life example: The shape of a school desk, a bucket.

3. Lines of Symmetry

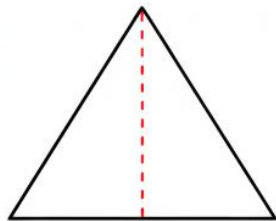
A line of symmetry is like a mirror line. If you fold a shape along this line, both halves match exactly. They are mirror images.



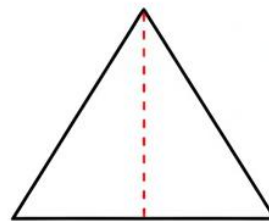
Square (4 Lines)



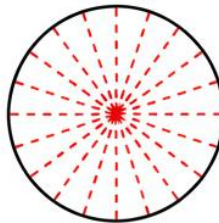
Rectangle (2 Lines)



**Equilateral Triangle
(3 Lines)**



**Isosceles Triangle
(1 Line)**



Circle (Many Lines)



- **Finding them:** Imagine folding the shape. If both parts sit perfectly on top of each other, the fold line is a line of symmetry.
- **Examples:**
 - A square has **4** lines of symmetry (through the middle of opposite sides and through the corners).
 - A rectangle has **2** lines of symmetry (through the middle of the longer and shorter sides).

- An equilateral triangle has **3** lines of symmetry.
- A circle has infinite lines of symmetry (any line through its center).

4. Congruence vs. Similarity

- **Congruent Shapes:**

- These shapes are identical twins. They are exactly the same shape and same size. If you cut one out, you could place it perfectly on top of the other with no part sticking out.
- Example: Two 10 Naira coins from the same year.

- **Similar Shapes:**

- These shapes are the same shape but different sizes. One is an **enlargement** or a **reduction** of the other.
- Example: A passport photograph and a poster-sized portrait of the same person. A small "Thank You" card and a large wedding invitation card of the same design.

EVALUATION

1. A triangle has sides measuring 5cm, 5cm, and 7cm. What type of triangle is it? Why?
2. How is a rhombus different from a square? Mention one key property.
3. Draw a simple trapezium and mark the pair of parallel sides with arrows.
4. How many lines of symmetry does an isosceles triangle have? Draw one and show the line.
5. True or False: All rectangles are squares. Explain your answer.
6. If you have a big map of Nigeria and a small map of Nigeria in your atlas, are they congruent or similar? Why?
7. Identify this shape: "I have four sides. My opposite sides are equal and parallel, but my angles are not 90° ."

ASSIGNMENT

1. **(Practical/Drawing)** Go on a "Shape Hunt" at home. Find and list one real-life object for:
 - a) Rectangle
 - b) Triangle
 - c) Square.
 Draw a simple sketch of each.

2. **(Written)** Draw an equilateral triangle of side 4cm (rough sketch is fine). Using dotted lines, show all its lines of symmetry. How many did you draw?
 3. **(Oral/Explanation)** Explain to a classmate, using your own words and examples, the difference between "congruent" and "similar."
-



WEEK 2: ANGLES

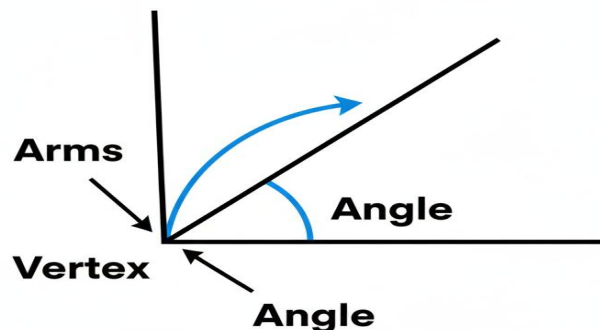
OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define an angle and identify types of angles (acute, right, obtuse, reflex).
2. Measure angles using a protractor.
3. Calculate missing angles on a straight line and at a point.
4. Find missing angles in a triangle.

CONTENT

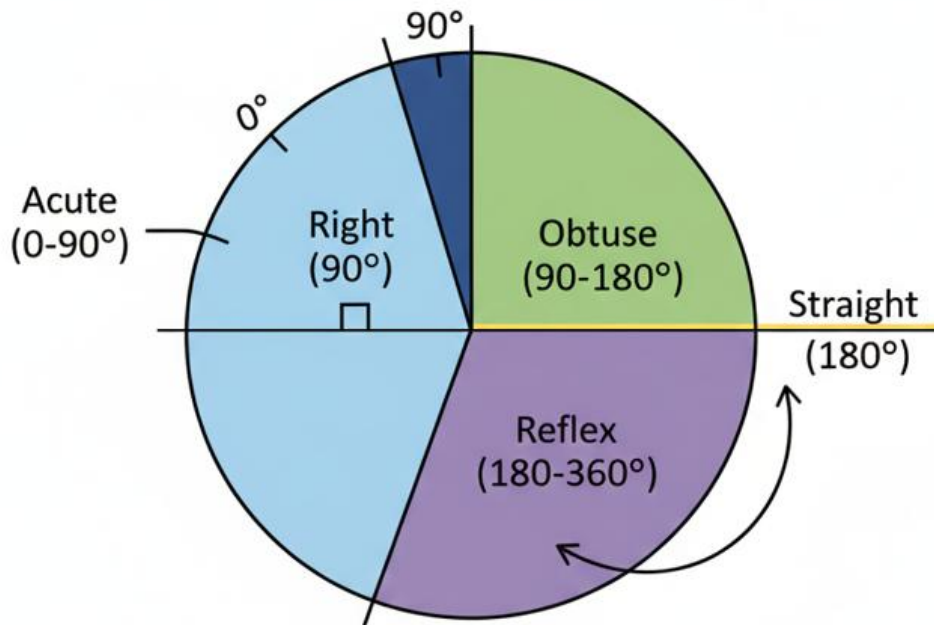
1. What is an Angle?

An angle is formed when two straight lines meet at a point. The point where they meet is called the **vertex**. The two lines are called **arms**. We measure the size of an angle in **degrees ($^{\circ}$)**, using a tool called a **protractor**.



2. Types of Angles

We classify angles by their size in degrees.



- **Acute Angle:** Less than 90° . It is small and sharp. Example: The angle between the hands of a clock at 2 o'clock.
- **Right Angle:** Exactly 90° . It forms a perfect "L" shape. Example: The corner of your mathematics textbook, the junction where two walls meet.
- **Obtuse Angle:** More than 90° but less than 180° . It is wider than a right angle but not a straight line. Example: The angle between the minute and hour hand at 4 o'clock.
- **Straight Angle:** Exactly 180° . It is a straight line. Example: A ruler lying flat on a desk.
- **Reflex Angle:** More than 180° but less than 360° . It is the larger angle when two lines meet. Example: The angle between the hands of a clock at 7 o'clock (the big angle, not the small one).

3. Measuring Angles with a Protractor

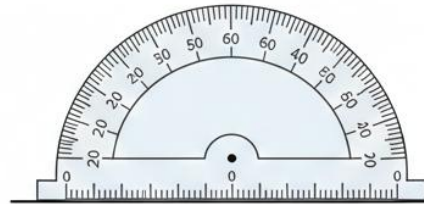
- **Step 1:** Place the **center mark** (small hole) of the protractor on the **vertex** of the angle.
- **Step 2:** Align the **baseline** of the protractor with **one arm** of the angle.
- **Step 3:** Read the number on the scale where the other arm passes. Be careful to use the correct scale (inner or outer).

How to measure a 60° angle

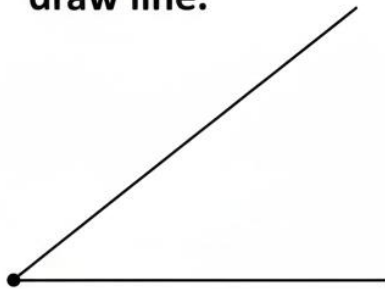
1. Draw a line.



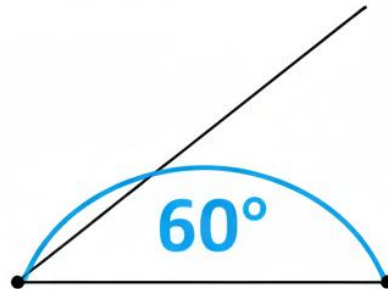
2. Place protractor & mark 60°.



3. Remove protractor & draw line.



4. Label the angle.



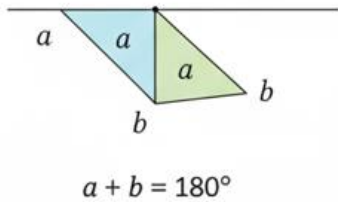
4. Key Angle Facts

These are rules that are always true. They are like mathematical laws.

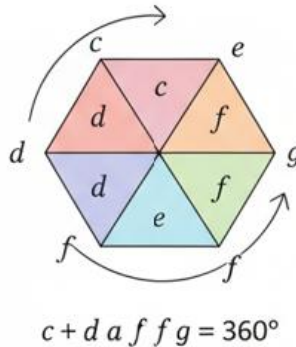
- **Angles on a Straight Line:** The sum of angles on a straight line is always **180°**.
 - If you know one angle, subtract it from 180° to find the other. Example: If one angle is 110°, the other is $180^\circ - 110^\circ = 70^\circ$.
- **Angles at a Point:** The sum of angles around a point is always **360°**.
 - Example: If you turn around completely, you have turned 360°.

- **Vertically Opposite Angles:** When two straight lines cross (like an X), the angles opposite each other are equal. They are called vertically opposite angles.

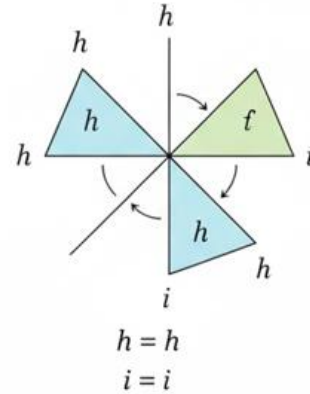
A. Angles on a Straight Line



B. Angles at a Point



C. Vertically Opposite Angles



5. Angles in a Triangle

This is a very important rule: The three interior angles inside **ANY** triangle always add up to **180°**. We can use this to find a missing angle.

- **Example:** A triangle has angles of 75° and 35° . What is the third angle?
 - Add the known angles: $75^\circ + 35^\circ = 110^\circ$.
 - Subtract from 180° : $180^\circ - 110^\circ = 70^\circ$.

EVALUATION

1. What is the name for an angle that is exactly 90° ?
2. An angle measures 150° . What type of angle is it?
3. Two angles sit on a straight line. One is 85° . What is the size of the other?
4. Find the missing angle x in this triangle: Angles are 50° , 60° , and x .
5. What is the sum of angles around a full point?
6. If two lines cross and one angle is 40° , what is the size of the vertically opposite angle?

ASSIGNMENT

1. **(Practical/Drawing)** Use a protractor to draw an angle of 120° . Label it as an obtuse angle.
 2. **(Written)** A triangle has one right angle (90°) and one angle of 45° . Calculate the size of the third angle. What type of triangle is this based on its angles?
 3. **(Investigation)** Look around your classroom. Find and sketch two examples of (a) a right angle, (b) an acute angle. Label the vertex and arms in one of your sketches.
-



WEEK 3: HEIGHTS AND DISTANCES

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Explain how to estimate height using the "Shadow Method".
2. Use the concept of scale to determine distances.
3. Solve simple problems involving similar triangles.

CONTENT

1. Indirect Measurement: The Shadow Method

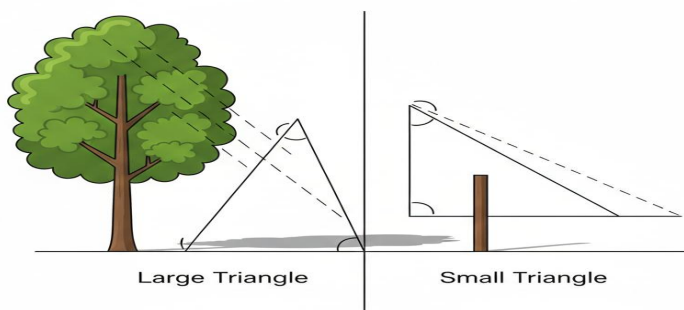
How can we measure a tall tree, a flagpole, or a building without climbing it? We use indirect methods. The **Shadow Method** is a common and clever one.

- **Why it works:** At the same time of day, the sun's rays are nearly parallel. This creates **similar triangles** between an object and its shadow.

Steps for the Shadow Method:

1. Measure the height of a short, vertical object you **can** measure directly (e.g., a metre ruler, a person).
2. Measure the length of its shadow on the ground.
3. At the exact same time, measure the length of the shadow cast by the tall object.
4. Set up a proportion and solve.

Similar Triangles



The sun's parallel rays create similar triangles, so we can use proportions to find unknown heights

- **Example:** A 1-metre stick casts a shadow of 0.5m. A tree casts a shadow of 6m.
 - Ratio of stick's height to its shadow: $1\text{m} / 0.5\text{m} = 2$.
 - This ratio is the same for the tree: $\text{Tree Height} / 6\text{m} = 2$.
 - Therefore, $\text{Tree Height} = 2 \times 6\text{m} = \mathbf{12\text{m}}$.

2. Understanding Similar Triangles

Similar triangles are triangles that have the same shape but different sizes. Their corresponding angles are equal, and the **ratios of their corresponding sides are equal**.

- The shadow method works because the triangle formed by the stick and its shadow is similar to the triangle formed by the tree and its shadow.

3. Scale and Map Reading

We cannot draw the real distance from Lagos to Abuja on a sheet of paper. A **scale** tells us how much smaller the drawing is compared to real life. It is a ratio.

- **Scale Format:** 1 cm : 10 km (This means 1 centimetre on the map represents 10 kilometres in real life).
- **How to use it:**
 - **To find REAL distance:** Measure the map distance in cm, then multiply by the "real life" part of the scale.
 - Example: On a map with scale 1cm : 50km, two towns are 4cm apart. Real distance = $4 \times 50 = \mathbf{200\text{km}}$.
 - **To find MAP distance:** Take the real distance, divide by the "real life" part of the scale.
 - Example: To represent 300km on the same map (1cm:50km), you would draw a line $300 \div 50 = \mathbf{6\text{cm}}$ long.



EVALUATION

1. Why can't we always use a tape measure to find the height of a tall building? What method can we use instead?
2. A man 1.8m tall casts a shadow 0.9m long. What is the height of a nearby pole that casts a shadow 3m long at the same time?
3. What tool do we use to measure distance on a flat map or drawing?
4. The scale on a map is 1cm to 2km. What is the actual distance if two villages are 7cm apart on the map?
5. Explain in one sentence why the shadow method works. (Hint: Think about the triangles).

ASSIGNMENT

1. **(Practical)** On a sunny day, measure your own shadow. Then measure the shadow of a tree or a pole in your school at the same time. If you are 1.4m tall, estimate the height of the tree using the shadow method. (Note: This is an in-class or take-home project).
2. **(Written)** The distance from Benin City to Asaba is about 100km. How long would a line representing this distance be on a map with a scale of 1cm : 25km?
3. **(Drawing)** Draw a simple sketch of your classroom from above (a plan). Use a scale of 1cm : 1m. If your classroom is 10m long, how many centimetres will you draw it?

WEEK 4: POLYGONS AND SCALE DRAWING

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define a polygon and name polygons with 5, 6, 7, and 8 sides.
2. Distinguish between regular and irregular polygons.
3. Calculate the sum of angles in a quadrilateral.
4. Interpret a simple scale drawing.

CONTENT

1. What is a Polygon?

A polygon is a flat, closed 2D shape made with three or more straight sides. The sides must **not** cross each other. We name polygons based on the number of sides they have.

Number of Sides		Name of Polygon	
3		Triangle	
4		Quadierital	
5		Pentagon	
7		Heptagon	
8		Octagon	
10	Decagon		

- **Pentagon (5 sides):** Example: The shape of a school pentagon house badge, the headquarters of the U.S. Department of Defense.
- **Hexagon (6 sides):** Example: The cells in a honeycomb, a nut.
- **Heptagon (7 sides):** Example: Some coins, like the British 50p coin (which is a curved heptagon).
- **Octagon (8 sides):** Example: A "STOP" road sign.

2. Regular vs. Irregular Polygons

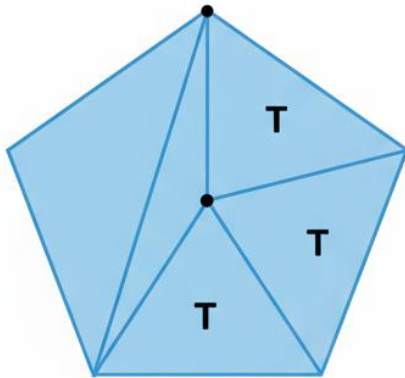
- **Regular Polygon:** All sides are equal in length, and all interior angles are equal in size. It is symmetrical.
 - Examples: An equilateral triangle, a square, a regular pentagon.
- **Irregular Polygon:** The sides are not all equal, and/or the angles are not all equal. It lacks symmetry.
 - Examples: A scalene triangle, a rectangle (unless it's a square), an irregular pentagon.

3. Sum of Interior Angles in a Polygon

We can find the total sum of all angles inside any polygon by splitting it into triangles. A simple formula exists: **Sum of Interior Angles** = $(n - 2) \times 180^\circ$, where **n** = number of sides.

- **Why?** A polygon with n sides can be divided into $(n - 2)$ triangles. Since each triangle has 180° , we multiply.
 - Quadrilateral (n=4): $(4-2) \times 180^\circ = 2 \times 180^\circ = 360^\circ$. (This is why angles in a rectangle add to 360°).
 - Pentagon (n=5): $(5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$.

Area of a Pentagon



Formula

Area of Pentagon = 3 x Area of one Triangle

Calculation

Area = 3 x (1/2 x base x height)

Area = 3 x (1/2 x 4cm x 3cm)

Area = 18 cm²

4. Scale Drawing in Real Life

Scale drawings are used by architects, builders, and engineers to plan houses, bridges, and machines before they are built. It allows them to fit everything on a piece of paper.

- **How to read one:** Always check the scale first (e.g., 1:100).
- **Application:** If a bedroom is drawn as 5cm long on a plan with scale 1:50, it means the real room is $5\text{cm} \times 50 = 250\text{cm}$ or **2.5m long**.

EVALUATION

1. What is the name given to a polygon with eight sides?
2. What is the key difference between a regular hexagon and an irregular hexagon?
3. Using the formula, calculate the sum of interior angles in a hexagon (6 sides).
4. A quadrilateral has three angles: 80° , 95° , and 70° . What is the size of the fourth angle? (Hint: Sum is 360°).
5. On a house plan with a scale of 1cm : 2m, a kitchen is drawn 3cm wide. How wide is the real kitchen?
6. How many triangles can a heptagon (7 sides) be divided into?

ASSIGNMENT

1. **(Drawing)** Draw a regular octagon (as best you can). How many lines of symmetry do you think it has? Draw them with dotted lines.
 2. **(Written Calculation)** A regular pentagon has five equal angles. Use the sum of angles formula to find the size of **EACH** interior angle in a regular pentagon. (Hint: Find the total sum first, then divide by 5).
 3. **(Practical)** Your teacher's table is about 2m long. Using a scale of 1cm : 0.5m, how long should you draw it in your book? Draw the line.
-

WEEK 5 & 6: MIDTERM EXAM AND BREAK



WEEK 7: EVERYDAY STATISTICS - POPULATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define population in statistics.
2. Interpret data presented in a population table or bar chart.
3. Explain the terms "Birth Rate" and "Migration".
4. Analyze a simple population pyramid.

CONTENT

1. What is Population in Statistics?

In mathematics and social studies, **population** means the total number of items or people in a group we are studying. It could be:

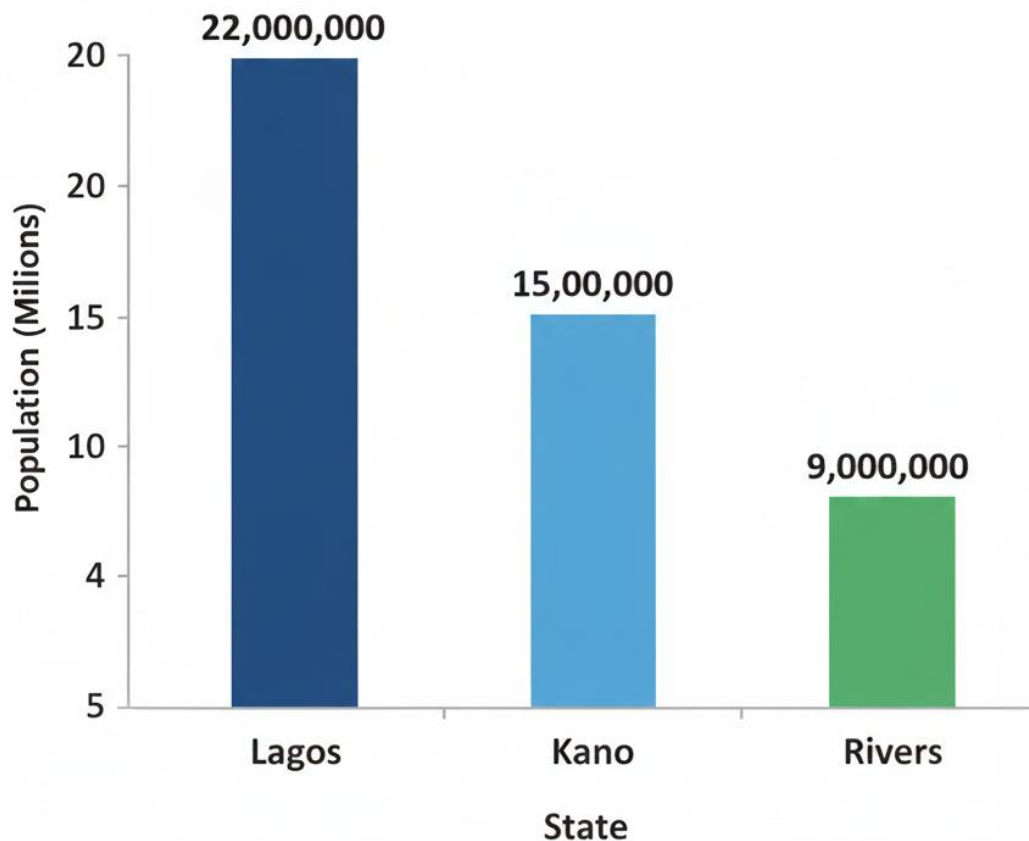
- The number of pupils in Primary 6.
- The number of cars in a park.
- The number of people in a country (like Nigeria's population).

2. Collecting and Organizing Population Data

Governments conduct a **census** to count the national population. This data is then organized so we can understand it easily.

- **Tables:** Present data in rows and columns. Clear and precise.
 - **Example Table:** Population of Some States (in millions)
 - Lagos: 12.5
 - Kano: 9.4
 - Rivers: 7.3
- **Bar Charts:** Present data using bars of different heights. Visual and easy to compare.
 - The taller the bar, the larger the population.

Population of Three States



3. Key Population Terms

- **Birth Rate:** The number of live births per 1,000 people in a year. A high birth rate means many babies are born, which can lead to population growth.
- **Death Rate:** The number of deaths per 1,000 people in a year. A high death rate can reduce population size.
- **Migration:** The movement of people from one place to another to live.
 - **Immigration:** Moving *into* a place (e.g., people moving to Lagos for work). Increases population.

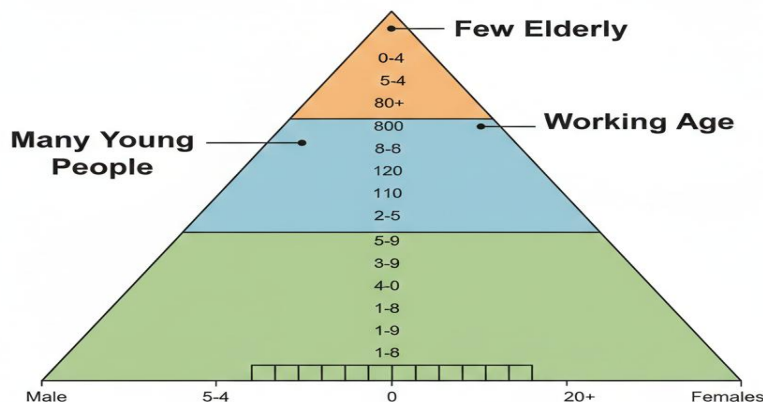
- **Emigration:** Moving *out* of a place (e.g., people leaving Nigeria to study abroad).
Decreases population.

4. The Population Pyramid

This is a special bar chart that shows the age and sex distribution of a country's population. It helps planners know how many young people, working adults, and elderly people there are.

- **Shape:** Often looks like a pyramid for developing countries.
- **How to read it:**
 - **Bottom (Base):** Shows young children (0-4 years, 5-9 years). A wide base means a high birth rate and many young people.
 - **Middle:** Shows working-age adults (15-64 years).
 - **Top (Apex):** Shows elderly people (65+ years). A narrow top means fewer people live to old age.
- **Why it matters:** A country with a wide base needs many schools and jobs in the future. A country with a wide middle has many workers.

Simplified Population Pyramid for Nigeria



EVALUATION

1. What is the main purpose of a national population census?
2. Looking at a table, how do you quickly identify which state has the largest population?
3. If a country has high immigration and a low birth rate, what might happen to its total population?
4. What does a "wide base" on a population pyramid tell us about that country?
5. From a bar chart showing school attendance, if the bar for "Primary 5" is the tallest, what can you conclude?
6. What is the difference between immigration and emigration?

ASSIGNMENT

1. **(Data Collection)** Conduct a mini-census in your own home. Create a table showing the number of people in these age groups: 0-12 (Children), 13-64 (Adults), 65+ (Elderly).
 2. **(Calculation/Interpretation)** Using your table from question 1, calculate the total number of people in your home. What fraction of them are children?
 3. **(Drawing)** Draw a simple bar chart to represent the number of pupils in each class from Primary 1 to Primary 6 in your school (ask your teacher for the numbers or estimate).
-

WEEK 8: MEASURES OF CENTRAL TENDENCY I (MEAN)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define the "Mean" or Average.
2. Calculate the mean of a set of raw data.
3. Solve word problems involving the mean.

CONTENT

1. What is the Mean?

The **Mean** is the most common type of average. It is a single number that represents the centre or typical value of a set of numbers. We use it to get a general idea.

- **Real-life use:** Finding the average score in a test, the average rainfall in a month, the average daily sales in a shop.

2. How to Calculate the Mean

It is a simple three-step process. Remember the formula: **Mean = (Sum of all values) ÷ (Number of values)**

- **Step 1 – ADD:** Find the sum of all the numbers in the data set.
- **Step 2 – COUNT:** Count how many numbers there are in the data set.
- **Step 3 – DIVIDE:** Divide the sum by the count.



3. Worked Examples

- **Example 1 (Test Scores):** Bola's scores in 5 maths tests are: 70, 85, 90, 60, 75.
 - Sum = 70 + 85 + 90 + 60 + 75 = **380**
 - Count = 5 scores

- $\text{Mean} = 380 \div 5 = 76$
- Bola's average score is **76**.
- **Example 2 (Word Problem):** A trader sold 8 bags of rice. The mean (average) weight of the bags was 50kg. What was the total weight of all the bags?
 - We know: $\text{Mean} = 50$, $\text{Count} = 8$.
 - Formula: $\text{Mean} = \text{Total} \div \text{Count}$, so **Total = Mean $\times$ Count**.
 - $\text{Total weight} = 50\text{kg} \times 8 = 400\text{kg}$.

4. Important Note about the Mean

The mean can be affected by very high or very low numbers in the data set.

- **Example:** The ages of five children are 4, 5, 5, 6, and 10. The mean age is $(4+5+5+6+10)/5 = 30/5 = 6$ years. The 10-year-old "pulled" the average up.

EVALUATION

1. In your own words, what does the "mean" tell us about a set of numbers?
2. Calculate the mean of these numbers: 15, 20, 25, 30, 10.
3. If the sum of 6 numbers is 300, what is their mean?
4. A pupil scored a mean of 72% in 5 subjects. What was her total percentage score? (Hint: Find the sum).
5. Find the average daily temperature if the temperatures for a week were: 32° , 34° , 33° , 35° , 34° , 36° , 30° .
6. Why might the mean not always be the best way to describe a "typical" value? (Think of the age example).

ASSIGNMENT

1. **(Written)** Find the mean of the first seven whole numbers: 1, 2, 3, 4, 5, 6, 7.
2. **(Word Problem)** If the mean weight of 4 boys is 40kg, what is their total weight? If a 5th boy weighing 48kg joins them, what is the new mean weight?
3. **(Practical/Oral)** Ask three friends for their height in cm. Calculate the mean height of your group. Share your steps.



WEEK 9: MEASURES OF CENTRAL TENDENCY II (MEDIAN, MODE, RANGE)

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Define and calculate the Median of a data set.
2. Define and identify the Mode of a data set.
3. Calculate the Range of a data set.
4. Distinguish between Mean, Median, and Mode.

CONTENT

1. The Median (The Middle Value)

The **median** is the middle number when a data set is arranged in order (from smallest to largest).

- **How to find it:**
 - a. **ARRANGE** the numbers in ascending order.
 - b. **FIND** the middle position.
- **For an ODD number of values:** The median is the single number in the middle.
 - **Example:** Scores: 12, 18, 13, 15, 20.
 - Arrange: 12, 13, 15, 18, 20.
 - Middle (3rd number): 15. **Median = 15.**
- **For an EVEN number of values:** The median is the **average** of the two middle numbers.
 - **Example:** Ages: 10, 7, 9, 12.
 - Arrange: 7, 9, 10, 12.
 - Two middle numbers: 9 and 10.
 - Median = $(9 + 10) \div 2 = 9.5$.

2. The Mode (The Most Frequent Value)

The **mode** is the value that appears most often (has the highest frequency) in a data set.

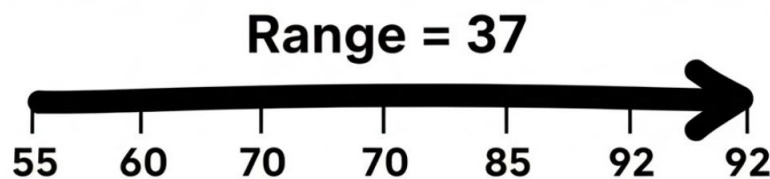
- **How to find it:** Count how many times each number appears. The one with the highest count is the mode.
 - **Example:** Shoe sizes sold: 6, 7, 8, 7, 9, 7, 8.

- Size 6 appears 1 time.
 - Size 7 appears 3 times.
 - Size 8 appears 2 times.
 - Size 9 appears 1 time.
 - **Mode = 7.**
- A set can have one mode, more than one mode (bi-modal), or no mode if all numbers appear once.

3. The Range (The Spread)

The **range** tells us how spread out the data is. It is the difference between the highest and the lowest values.

- **Formula: Range = Highest Value - Lowest Value**
- **Example:** Test scores: 55, 92, 70, 85, 60.
 - Highest = 92, Lowest = 55.
 - Range = $92 - 55 = 37$. This shows a wide spread in scores.



4. Comparing Mean, Median, and Mode

- **Mean:** Uses all data; can be affected by extreme values.
- **Median:** Not affected by extreme values; gives the true middle.
- **Mode:** Shows the most popular or common item.
- **Real-life choice:** A shoe shop owner cares about the mode (most popular size). A teacher looking at typical performance might prefer the median if one pupil scored zero.

EVALUATION

1. What is the most important first step when finding the median?

2. Find the median of: 21, 15, 18, 25, 20.
3. Find the mode of: Red, Blue, Red, Green, Blue, Blue, Yellow.
4. Calculate the range of: 100, 55, 80, 95, 62.
5. In a survey of favourite fruits, Orange got 10 votes, Mango 15, and Banana 15. What is the mode?
6. Why might the median be a better average than the mean for reporting house prices in an area?

ASSIGNMENT

1. **(Written)** The heights (in cm) of five plants are: 12, 18, 14, 16, 18. Find the (a) Median, (b) Mode, (c) Range.
2. **(Data Analysis)** Record the number of siblings (brothers and sisters) each person in your group has. Find the mode of this data. What does it tell you?
3. **(Oral/Explanation)** A class has test scores: 10, 10, 10, 90, 90. The mean is 42. Explain why the mean might be misleading here. What other average would be useful?

WEEK 10: COMPREHENSIVE PROBLEM SOLVING & SECONDARY SCHOOL PREPARATION

OBJECTIVES By the end of the lesson, pupils should be able to:

1. Solve multi-step word problems involving basic operations.
2. Apply mathematical concepts to real-life situations (shopping, time, measurement).
3. Demonstrate readiness for secondary school mathematics through critical thinking questions.

CONTENT

1. Tackling Multi-Step Word Problems

These problems require you to do more than one calculation. The key is to read carefully, plan your steps, and show your working.

- **Strategy:**
 - a. **Understand:** What is the question asking for? Underline key numbers and words.
 - b. **Plan:** Decide what operations (+, -, ×, ÷) you need and in what order.
 - c. **Solve:** Perform the calculations step-by-step.
 - d. **Check:** Does your answer make sense? Reread the problem.
- **Example:** "Musa bought 3 exercise books at ₦250 each and 2 pens at ₦150 each. He paid with a ₦2000 note. How much change did he receive?"
 - Step 1 (Cost of books): $3 \times \text{₦}250 = \text{₦}750$
 - Step 2 (Cost of pens): $2 \times \text{₦}150 = \text{₦}300$
 - Step 3 (Total cost): $\text{₦}750 + \text{₦}300 = \text{₦}1050$
 - Step 4 (Change): $\text{₦}2000 - \text{₦}1050 = \text{₦}950$

2. Critical Thinking and Logic Puzzles

Mathematics is not just about numbers; it's about thinking logically.

- **Example 1:** "If you have a single matchstick and enter a dark room with a kerosene lamp, an oil heater, and a candle, what do you light first?"

- Answer: The **matchstick**. (This tests if you follow the sequence of events).
- **Example 2 (Math Logic):** "A farmer has 17 sheep. All but 9 run away. How many are left?"
 - Answer: **9 sheep**. ("All but 9" means everything except 9 ran away, so 9 remain).

3. A Glimpse into Secondary School Mathematics (JSS1)

In Junior Secondary School, you will build on what you know.

- **Algebra:** You'll use letters for unknowns more formally. If 3 more than a number is 11, we write: $n + 3 = 11$, then solve $n = 8$.
- **Integers:** You'll learn about negative numbers (e.g., -5, -10). They are used for temperatures below zero, debts, or floors below ground level.
- **Geometry:** You'll study angles and shapes in more detail.
- **Key Advice for Success:**
 - Show all working. Marks are given for correct steps.
 - Ask questions when you don't understand.
 - Practice regularly. Mathematics improves with practice.

EVALUATION

1. A bag of rice costs ₦32,000. A woman buys $\frac{1}{2}$ bag. How much does she pay?
2. A car travels 75km in 1 hour. How long will it take to travel 300km at the same speed?
3. Calculate the perimeter of a square with an area of 64 cm^2 . (Hint: Find the side length first).
4. Solve using BODMAS: $12 + (8 \div 2) \times 3$.
5. Logical Thinking: If two hours ago, it was as long after 1 o'clock as it was before 1 o'clock, what time is it now? (Hint: Think about symmetry around 1 o'clock).
6. A packet contains 24 sweets. They are shared between A and B in the ratio 3:5. How many sweets does B get?

ASSIGNMENT

1. **(Multi-Step Problem)** Mr. Bello's monthly salary is ₦120,000. He spends $\frac{1}{3}$ on rent and $\frac{1}{4}$ on food. How much money does he have left for other things?
 2. **(Measurement & Logic)** A rectangular garden is 15m long. Its perimeter is 50m. What is its width? What is its area?
 3. **(Preparation Task)** Interview a JSS1 or JSS2 student in your community. Ask them: "What is one new: in mathematics you learned this year that you didn't know in Primary 6?" Write down the: name and one thing about it.
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